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The development of wishes in child Greek

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Abstract

This thesis investigates the development of desires and wishes in child Greek.

It attempts to shed light on whether children distinguish between attainable and unattainable desires (O-marked and X-marked respectively in von Fintel & Iatridou's (2023) terms), the age at which we observe a difference in the developmental path, and, ultimately, the age at which children have acquired the necessary grammatical cues that denote counterfactuality. A comprehension picture-matching task, in a 'Tell-me-who-is-talking' mode was run on 30 typically developing children (aged from 4-9 years old), divided into two age groups (mean age: 6;4 and 8;9) and 30 adults (control group; mean age 34).

According to the experimental procedure, children were told a short story about little creatures who went to the Festival of Desires. During the Festival, the little creatures can become any color they want. Then children are informed that they will listen to the 'interviews' the little creatures gave, talking about the color they got and if they want to remain the same or if they would like to be different. Children would watch two little creatures on the screen, while listening to an utterance, and they had to tell the experimenter which creature is talking and click on the icon.

Overall, the findings suggest that younger children and a subgroup of the older children interpret past X-marked desires as O-marked, due to Actuality bias (de Villiers 2005; Kazanina et al 2020) and possibly because of the misanalysis of the pluperfect associated with X-marking (Crutchley 2004; Rouvoli et al 2019; Tulling & Cournane 2019, 2022). It is concluded that X-marking interpretation of desires is a prolonged developmental process. The findings are in accordance with previous findings on the development of counterfactual reasoning (Beck et al. 2006; Rafetseder & Perner 2011; Rafetseder et al 2013), claiming that the development of counterfactual conditionals is not completed by the age of 12.

Keywords

desires, wishes, X-marking, counterfactuality, child language development, language acquisition, counterfactual reasoning

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Contents

List of abbreviations and logical symbols

Abbreviations used in glosses

1/2/3PL	first/second/third person plural
1/2/3SG	first/second/third person singular
ACC	accusative
AUX	auxiliary
COND	conditional mood
F	feminine
FUT	future marker
HAB	habitual aspect
IPFV	imperfective aspect
IND	indicative mood
INTJ	interjection
NA	the element *na* in Modern Greek
NPST	non-past tense
PERF	perfective aspect
PLUP	pluperfect
PST	past tense
SUBJ	subjunctive mood

Logical symbols

$\wedge$	conjunction ('and')
$\forall$	universal quantifier ('for all/ every')
$\in$	set membership ('it belongs to/ is an element of')
$\equiv$	identical to
$\not\equiv$	not identical to
$\vDash$	semantically entails
$\nvDash$	does not semantically entail
BEST(w,x)	best worlds according to (the desires of) the individual **x** in the world **w**
BOUL	bouletic alternatives/ desire set
D	the domain **D** is the domain of desire ascriptions that is related to or identical to the set of doxastically accessible worlds for the agent of the desire, the agent's belief set or doxastic set
DOX	doxastic alternatives/ belief set
$Sim_{a(b)}$	function that shows the similarity of **a** to **b**
p(w')=1 w'	world **w'** where the proposition **p** is true

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Introduction

This thesis explores the development of desires and wishes in child Greek. According to von Fintel & Iatridou (2023) desires that are open to fulfillment are O-marked, whereas desires that cannot be fulfilled in the actual world, or their truth is a yet unknown (X) issue, are X-marked. It is thus suggested that all wishes that are counterfactual/ unattainable in the actual world are X-marked desires.

In the first chapter, I provide a definition and brief description of desires and wishes. In the second chapter, I focus on the linguistic components that make a wish a wish. First, I refer to the similarity of wishes to counterfactual conditionals showing how wishes are formed. A crosslinguistic overview of wishes, shows that languages can express wishes mainly in two different ways regarding the form of the desire predicate. After presenting the form of wishes in different languages, I describe the meaning of *want* and *wish*-sentences towards understanding the semantics of desire.

In the third chapter, I present the morphosyntactic components of X-marked desires in Greek, focusing on the subcategorization information of the main desire predicates, on the use of past tense and imperfective aspect morphology, as well as on the role of optativity.

In the fourth chapter, I offer an overview of previous studies that are related to the development of wishes. The overview starts with a description of the naturalistic studies on counterfactual wishes and conditionals in child English, based on spontaneous child speech from CHILDES, by Tulling and Cournane (2019, 2022). It then focuses on studies on the development of counterfactual structures, counterfactual reasoning and counterfactual emotions that are related to the development of wishes (de Villiers 2005; Beck et al. 2006; Rafetseder et al. 2010, 2013; Rafetseder & Perner 2010, 2011, 2014, Beck & Rigs 2014; Crutchley 2004; Rouvoli et al. 2019; Kazanina et al. 2020).

In the fifth chapter, I present the comprehension experiment designed to explore the acquisition of X-marked desires in Modern Greek and the findings.

Finally, in the sixth chapter I discuss the issues raised by the results of the study and the question that remain open. The references and the appendices with the materials used in the experiment of this study follow at the end of this thesis.

1 Wishes as unattainable desires

1.1 What are desires

Desires are attitudes towards an imaginative end or an object that drives the purpose of any action (Russell 1921). We can desire water, food, sleep, love, communication, but also knowledge, travelling, storytelling, enjoyment of art and expression through it. Epicurus (Letter to Meoneceus 127-8) classifies desires into three types. Necessary and natural desires are in harmony with life itself and aim to happiness. Merely natural desires aim at satisfaction from pleasurable activities and things. Finally, empty desires, like immortality, fame and wealth are incompatible with genuine needs (Konstan 2022).

We tend to think of the object of our desire as the object of our belief. The attitude towards the object of desires is what makes it different form the object of belief that differentiates the desire from the belief. If desires were beliefs, we would desire what we believe to be good. In the actual world though, we can believe in things like the benefits of love and knowledge and have no desire for them (Lewis 1996). We can also imagine scenarios in which what we believe to be good contradicts what we want (Lewis 1988). For every desire, there is the desire itself and the attitude of desiring it. To desire is to be in an emotional state that affects our feelings, our thoughts, and our actions (Schroeder 2006, 2020).

If all ethical restrictions were removed, then we would not think of desires as beliefs. Russell (1921) uses the example of hunger, to claim that the traits of an animal's bodily behavior show its desires and not its mental state. Animals have instincts towards the movements that help them get out of their discomfort, i.e. to stop their hunger, build their nests, protect their babies, and they are not thinking of the cycle of actions that need to be taken to survive as a species.

Both animal and human desires are expressed though bodily behavior traits. From a psychoanalytical perspective, Freud presents desire as a continuous need for satisfaction that is related to libido, whereas Lacan describes desire as absence that stems from the subject's alienation from its mother during birth (Kulick 2003). When a child is born, it is separated from its mother, and it experiences a lack of completeness. Lacan defines desire as the desire of the Other, that is "produced in the margin which exists between the demand for the satisfaction of need and the demand for love" (Lacan 2011; 404). Furthermore, he suggests that the object of desire is the place of not feeling complete, otherwise described as the position of the subject in relation to the other. (Lacan 1978; Lacan & Gallagher 2002; Lacan 2011 (1957-1958 seminars)).

Language is one of the tools humans use to convey their desires, starting from infants. Through language, speakers express their "own conceptualization of reality in the attempt to structure possibilities according to their knowledge, beliefs, memories, expectations, desires, and priorities. Across modal verbs, adverbs, and propositional attitude verbs, language reveals that humans anchor reality not only to truth but to their own subjective understanding of truth" (Giannakidou & Mari 2021).

1.2 What are wishes

To wish is to express a desire and desires can be attainable or unattainable. In language, unattainable desires in the actual world have been described as *X-marked desires* by von Fintel & Iatridou (2023). The actual world is a member of the set of possible worlds, and it includes us and our surroundings however remote in time and space (Lewis 1986).

Expressing an unattainable/ counterfactual desire through wishing requires the ability to think counterfactually. The expression of counterfactual wishes in language reflects this ability through morphology and syntax (Iatridou 2000). X-marking is the term used to describe the morphological marking that is used in counterfactual wishes, as well as in counterfactual conditionals, and counterfactual necessity structures (von Fintel & Iatridou 2023).

When a desire is open to fulfillment, and its truth is an open (O) issue, it is O-marked (1a), whereas when a desire cannot be fulfilled in the actual world or its truth is a yet unknown (X) issue, it is X-marked (1b).

(1) a. I want to plant a carob tree in the yard.
  b. I wish I had planted a carob tree in the yard.

All wishes that are unattainable in the actual world and thus counterfactual, are X-marked desires. Not all X-marked desires are counterfactual though. X-marking on desires is ambiguous. In (1b), the attitude holder did not plant a carob tree in the yard. The desire in this case is X-marked and counterfactual and there is no shift in the evaluation world. In (2) the attitude holder expresses a desire that would be attainable in a world where the yard would be bigger and even though this world is counterfactual, the desire is not. In this scenario the desire is X-marked, but not counterfactual and there is a shift in the evaluation world. X-marked desires that are unattainable in the actual world are endo-X (1b). X-marked desires that are attainable in a counterfactual scenario are exo-X (2).

(2) If the yard was bigger, I would want to plant a carob tree.

X-marking does not encode counterfactuality. Similarly to exo-X desires, there are counterfactual conditionals that are X-marked, but do not have a counterfactual meaning (3).

(3) a. If you took the 5pm train you would get there by midnight.
  b. If Jones had taken arsenic, he would show exactly the symptoms he is in fact showing.
  c. There were no muddy footprints in the parlor, but if the gardener had done it, there would have been muddy footprints in the parlor, so the gardener must not have done it.

(examples from von Fintel & Iatridou 2023: pp. 1468, 1479, 1480)

In (3) the antecedent of all three conditionals is X-marked but not counterfactual as it is possible that *you can still take the 5pm train* (3a), that *Jones did take arsenic* (3b) and *that the gardener did not do it* (3c). These are the Future Less Vivid (FLV), Andersontype and Modus Tollens cases, respectively. These sentences have a counterfactual structure, but not a counterfactual meaning, as the counterfactuality that seems to arise with the antecedent, is cancelled by the consequent. This serves as an argument to the view that counterfactuality is a conversational implicature and not an assertion or a presupposition[1] (Iatridou 2000).

[1] An implicature is what is taken to be meant without being straightforwardly said. Two features of conversational implicatures obey the maxims of conversation (Grice 1975) and can be cancelled. A presupposition is what is taken for granted when something is said (Stalnaker 1973). An assertion is what the speaker says and believes to be true, that is "the expression of the world as being a certain way" (Stalnaker 1978).

2 What makes a wish a wish

2.1 Similarities between wishes and counterfactual conditionals

Across languages, there is a morphological similarity between X-marked desires and X-marked conditionals, formulated as the Conditional/ Wish pattern (4) (von Fintel & Iatridou 2023). X-marked conditionals follow the pattern “if p, then q”, where p has antecedent X-marking morphology and q has consequent X-marking morphology. X-marked desires follow the pattern “I want that p”, where p has antecedent X-marking morphology, and the consequent X-marking morphology appears on the desire predicate.

(4) The Conditional/ Wish pattern
X-marked conditional: if $p_{\text{ANTECEDENT}}$ then $q_{\text{CONSEQUENT}}$
X-marked desire: I $want_{\text{CONSEQUENT}}$ that $p_{\text{ANTECEDENT}}$
(von Fintel & Iatridou 2023: 1485)

If the desire verb is X-marked (5a), it has the same morphology with the verb of the consequent of the conditional (5b). Similarly, the complement of the desire-verb in X-marked desires (5a) and the antecedent of an X-marked conditional (5b), appear with the same morphology. The complement of desire-verbs in X-marked desires and the antecedent of counterfactual conditionals appear with the same morphology (Iatridou 2000). Pluperfect is used in the complement of the desire predicate in (5a) and in the antecedent of the conditional in (5b), indicating that both structures are X-marked. Moreover, the morphology of the consequent of the conditional in (5b) is identical to that of the desire predicate in (5a). In both cases, there is past imperfective.

(5) a. Tha ithela na ixa fitepsi ena dendro stin avli.
MOD want.1SG.PST,IPFV NA plant.1SG,PLUP a tree in-the yard
‘I wish I had planted a tree in the yard.’

b. An ixa fitepsi ena dendro stin avli, tha imun xarumeni.
If plant.1SG,PLUP a tree in-the yard MOD be.1SG.PST,IPFV happy
If I had planted a tree in the yard, I would be happy.’

Languages vary with respect to how they express wishes. Some languages express X-marking through combinatory morphology on the desire predicate and they are called transparent wish languages. Other languages express X-marking through a wish-dedicated word and they are not wish transparent. However, there is no clear distinction between two types of languages as regards wishes. Languages can express wishes both transparently and non-transparently. In what follows, I present examples of wish expression in languages that can be transparent or non-transparent but not necessarily only the one or the other type.

2.2 Wishes across languages

In languages wishes can be expressed transparently or non-transparently. This distinction is based on how the verb 'wish' is expressed in English (von Fintel & Iatridou 2023). In transparent wishes the meaning of wish is expressed by X-marking morphology on the desire predicate. In non-transparent wishes, the meaning of wish is expressed by a wish dedicated word.

In Greek, wishes are expressed either by X-marking morphology on the desire predicate or by a wish dedicated word, that is, both transparently and non-transparently, suggesting that languages are not strictly classified into one category or another. The form of X-marked desires in Modern Greek is presented in Chapter 3.

2.2.1 Transparent wishes languages

In transparent wish languages there is consequent X-marking morphology on the desire predicate and antecedent X-marking morphology on the complement of the desire-verb. Greek, Spanish, Farsi, Hungarian and (probably) Tugun are some languages that satisfy this pattern fully. In languages like Spanish, there are morphological differences between the antecedent and the consequent X-marking, whereas in other languages, like Hungarian, there are no morphological differences.

In X-marked desires in Spanish, consequent X-marking appears with conditional mood on the desire verb *querer* ('I want) and antecedent X-marking appears with past subjunctive on the complement of *querria (*'I wish'*)* (6).

(6) Querria que fuera más alto de lo que es. SPANISH
want.1SG,COND that be.3SG,PST,SUBJ more tall than it that it
'I wish s/he was taller than what s/he is.'
(von Fintel & Iatridou 2023: 1487)

In Farsi, X-marked desires are formed with X-marking morphology on the desire predicate and past imperfective (7a) or pluperfect (7b) in the complement (Mirazzi 2022).

(7) a. del-am mi-xast mi-am-ad FARSI
heart-my IPFV.want.PST,3SG IMPERF.come. PST,3SG
'I wish she had come'.

b. del-am mi-xast umade bud FARSI
heart-my IPFV.want.PST,3SG come.PP AUX.PST-3SG
'I wish she had come'.
(Mirrazi 2022: 244-246)

In Greek, Spanish and Farsi, the desire verb used in X-marked desires, is a verb meaning 'I want'. There are languages, like Tugun and Hungarian, in which the desire predicate employed has the meaning of 'I like' (Hungarian (8)) or 'I take' (Tugun (9)). X-marking morphology on these cases conveys the meaning of 'wish'.

In Hungarian, both antecedent and consequent X-marking appear with the X-marker *-nA*. Wishes are formed with the verb *szeretem* ('I like') and its complement that contains the dedicated marker *-nA,* as expected (8).

(8) Szeretném ha Márcsi tudna a választ. HUNGARIAN
like-nA-1SG if Marcsi know-3SG-nA the answer.ACC
'I wish Marcso knew the answer.'
(von Fintel & Iatridou 2023: 1486-7)

In Tugun (9), X-marked desires are formed with the verb 'marrala', that is the verb 'I take' combined with an irrealis morpheme[2].

(9) Marr-ala a hira. TUGUN
3PL.IRREALIS-take for them
'They wanted to take them for themselves.'
(Hinton 1991: 86)

There are languages that express counterfactuality through morphemes that are 'equivalent to the past tense but from other domains' (Karawani 2014: 92). In Burmese, counterfactuality (and X-marking as described here) is formed with the spatial morpheme *khé* ('in another place'). In addition to that, the irrealis marker *-me* is used in consequent X-marking, predicting that it will be part of the desire predicate morphology in X-marked desires (10).

(10) shèi θau? khé yin, nei kimn la gé léin-me. BURMESE
medicine drink KHE if, stay good come KHE predictive-irrealis
'If he took the medicine, he would have gotten better.'
(from a talk by Nichols 2002 in Nevins 2002: 442)

The temporal (6-9) and spatial (10) morphemes mentioned are used to give access to alternative worlds. Such morphemes are used in counterfactual desires and conditionals and thus in X-marked structures.

2.2.2 Non-transparent wish languages

In non-transparent wish languages, the meaning of 'I wish' is expressed by a phrase that has a wish meaning and not by X-marking morphology on the desire predicate. English, Turkish, Hindi, Croatian and (partly) Greek are among the languages that can express wishes non-transparently. It should again be noted that the categorization of languages as transparent and non-transparent is not strict, meaning that there are languages that can express wishes in both ways.

[2] The data from Tugun were found in Hinton (1991: 86-88), through *Feature 124A: 'Want' Complement Subjects* from the World Atlas of Language Structures (WALS). The fact that the desire is described as past and that the morpheme that makes it past is a morpheme that expresses irrealis modality, led to the description of this sentence as X-marked. It is not mentioned that the desire is counterfactual. More data are needed to confirm whether X-marked desires in Tugun are indeed formed as described here.

In English, X-marked desires have the form *'I wish that p* $_{ANTECEDENT}$*'* (11a). However, Longebaugh (2019) shows that there are transparent wishes in English too (von Fintel & Iatridou 2023: 1487, footnote 35). (11b) conveys that the speaker did not plant a carob tree in the yard. It also conveys that in all cases in which the speaker plants that tree in the yard, the speaker likes that fact. In other words, even though the desire in (11a, b) is X-marked, Longebaugh suggests that X-marked desires, like that in (11b), express an actual world desire towards a counterfactual scenario. He further suggests that a conditional form like that in (11b) is a non-logical if construction as it behaves like the *that CP* in the CP-linking environment, and not like a conditional adjunct (Longebaugh 2019: 103-111, 127-133)

(11) a. I wish I had planted a carob tree in the yard. ENGLISH
b. I would like it if I had planted a carob tree in the yard.

In Turkish, X-marked conditionals contain the conditional affix -*sa* and past tense in antecedent X-marking and aorist and past in consequent X-marking (12a). In X-marked desires *keşke* ('I wish') is combined with the -*sa* morpheme and past tense morphology and only the antecedent X-marking part is satisfied (12b). The order of the morphemes on the verb shows whether it is X-marked or O-marked. When the conditional -sa precedes past tense, the sentence is X-marked (12c), whereas when past precedes -sa, the sentence is O-marked (12d).

(12) a. John önümüzdeki salı gel-se-ydi, annesi çok mutlu ol-ur-du. TURKISH
John next Tue come.SA,PST his.mom very happy be-AOR-PST
'If John arrived next Tuesday, his mom would be very happy.'

b. Keşke önümüzdeki salı gel-se-ydi.
Keşke next Tuesday come.SA,PST
'I wish he would come next Tuesday'.

(von Fintel & Iatridou 2023: 1488)

c. Abelard Eloisee dün çiçek ver-se-ydi...
Abelard Eloise yesterday flowers give. SA,PST
'If Abelard had given flowers to Eloise yesterday...'

d. Abelard Eloisee dün çiçek ver-di-yse...
Abelard Eloise yesterday flowers give.PST,SA
'If Abelard gave Eloise flowers yesterday...'

(Nevins 2002:446)

In Hindi, X-marked conditionals appear with progressive and habitual aspect both in antecedent and consequent. In X-marked desires, *kaash* ('I wish') is combined either with progressive and habitual aspect or only with habitual (13).

(13) Kaash vo gaa rahaa ho-taa. HINDI
Kaash he sing.PROG be-HAB
'I wish he was singing.'

(von Fintel & Iatridou 2023: 1489)

In Croatian,[3] X-marked conditionals are formed with past tense in antecedent X-marking and past tense and aorist in consequent X-marking (14). X-marked desires are formed with *bar(em)* ('at least') and antecedent X-marking morphology on the verb (15a). An alternative way of expressing an X-marked desire in Croatian is with the verb *wish* in past tense, a CP complement in past and aorist (15b). An X-marked desire like (15b) is acceptable in a context about past regrets. Otherwise, it feels unnatural. The sentence '*I wish I had planted a tree in the yard*' can be either expressed as 'If at least I had planted a tree in the yard' (15a) or *'I would wish I had planted a tree in the yard'* (15b).

(14) Da sam posadila drvo u dvorištu, bila bih jako sretna. CROATIAN
that be.1SG plant.PST.1SG.F tree in yard be.PST.1SG.F be.AOR very happy
'If I had planted a tree in the yard, I would be very happy.'

(15) a. Da sam barem posadila drvo u dvorištu.
that be.1SG at-least plant.PST.1SG.F tree in yard
'I wish I had planted a tree in the yard.'

b. Željela bih da sam posadila drvo u dvorištu.
wish.PST.1SG.F be.AOR that be.1SG plant.PST.1SG.F tree in yard
'I would wish I had planted a tree in the yard.'

(Iva Kovač, p.c.)

Summarizing the crosslinguistic overview of X-marked desires, languages vary depending on whether they express wishes with consequent X-marking morphology on the desire predicate or with a 'wish' dedicated verb or phrase and antecedent X-marking morphology on the complement.

Understanding how desires and wishes are expressed in language does not only require knowing about the morphological expression of O- and X-marking, but also being able to describe the meaning of want and wish sentences. In the following section I describe the semantics of desires and wishes.

2.3 The semantics of want and wish

2.3.1 Modality of desire

Modality refers to how language expresses the relationship of propositions to truth. Main clauses are typically evaluated in the actual world. Within the possible world semantic theory, modals quantify over possible worlds. The various readings of modality are called modal flavors and they differ "in what kind of facts in the evaluation world they are sensitive to" (Heim & von Fintel 2011). The primary distinction is between epistemic/doxastic modals which quantify over worlds consistent with the evidence or the beliefs of an individual and circumstantial modals, which are further differentiated depending on whether they express requirements (deontic modality), goals (teleological

[3] I would like to thank Iva Kovač for the examples in Croatian.

modality) and desires (bouletic modality) (Kratzer 1981). In the following lines, I discuss desire reports as they are the ones used as experimental items in this study.

2.3.2 Desire reports

Propositional attitude reports are sentences with psychological verbs (such as *believe, dream, imagine, (be) happy, wish, hope, intend*) which have the property of embedding finite or non-finite clauses. There are two types of desire reports: the propositional desire reports that include a CP complement (16a) and the desire reports that include a DP complement (16b). In (16b) *want* is an intensional transitive verb, as it displays properties that are characteristic of propositional attitude reports, like (16a), but not of transitive verbs like *catch* in (16c) (Grano 2021: 1,198).

(16) a. Harry wants to buy an owl. PSYCHOLOGICAL VERB
b. Harry wants an owl. INTENSIONAL TRANSITIVE VERB
c. Harry caught an owl. TRANSITIVE VERB

The *de re* – *de dicto* ambiguity, which is a property of all intensional predicates, is also a main property of desire reports (16a, b). The *de re* ('of the thing') reading refers to an item of the actual world that remains the same inside and outside of the intensional context, whereas the *de dicto* ('of what is said') reading refers to an item that is different inside and outside of the intensional context depending on its description (Quine 1960). The *de re* reading of (16a) and (16b) is that there is a specific owl in the actual world that Harry wants (to buy). The *de dicto* reading suggests that every possible world where Harry fulfills his desires, is a world where he owns or buys an owl.

The second intensional property of *want*-sentences is that their *de dicto* reading does not presuppose the existence of an owl (17a, b). In a sentence like (16c), however, the existence of an owl is presupposed, and ungrammaticality occurs if it is interpreted as *de dicto* (17c).

(17) a. Harry wants to buy an owl, but there is no owl.
b. Harry wants an owl, but there is no owl.
c. *Harry caught an owl, but there is no owl.

The third property of propositional desire reports is that they do not accept the substitution of extensionally equivalent phrases (18). Extensionally equivalent phrases are phrases that can be equal in reference, but not in sense, in that they express different modes of presentation[4].

(18) a. Harry wanted to catch Superman. ≢ Harry wanted to catch Clark Kent.
b. Harry wanted Superman. ≢ Harry wanted Clark Kent.
c. Harry caught Superman. ≡ Harry caught Clark Kent.

[4] The sense and reference distinction firstly appeared in Frege (1892) as 'Sinn und Bedeutung'.

Superman and Clark Kent are extensionally equivalent phrases, as they are the same person, but they differ in the way they are presented to the world. If Harry wants (to catch) Superman, it is not necessary that he wants (to catch) Clark Kent (18a, b), whereas if he catches Superman, it is necessary that he has caught Clark Kent (18c).

2.3.3 The semantics of want-sentences

Desire reports are clause embedding *want*-sentences. A possible semantic representation of desire reports would be similar to that of belief reports. That is, universal quantification over bouletic alternatives for *want*-sentences as it is used over doxastic alternatives for *believe*-sentences [5] (19).

(19) a. $[[\text{Beatrix believes it's raining}]]^{W}$ $\forall w' \in DOX_{b,w}$: rain(w')
where $DOX_{x,w} = \{w' \mid w' \text{ is compatible with x's beliefs in w}\}$

b. $[[\text{Beatrix wants it to be raining}]]^{W}$ $\forall w' \in BOUL_{b,w}$: rain(w')
where $BOUL_{x,w} = \{w' \mid w' \text{ is compatible with x's desires in w}\}$

(Grano 2021:154)

'Beatrix believes it's raining' means that in all worlds that are compatible with Beatrix's beliefs, it is raining (19a). In the same way, 'Beatrix wants it to be raining' means that in all worlds that are compatible with Beatrix's desires, it is raining (19b).

Heim (1992) suggested that the semantic analysis of *believe*-sentences is too strong to be used with *want*-sentences and thus (19b) is wrong. Her argument was that in a sentence like (20), her desire to teach on Tuesdays and Thursdays comes in contrast with the worlds that are compatible with everything that she desires, because in these worlds, she does not teach at all.

(20) I want to teach Tuesdays and Thursdays next semester.

(example from Heim 1992: 195)

The meaning of *want* can be described through the relationship between an individual x, the evaluation world, and the set of worlds that x considers as possible and desirable. In other words, *want* expresses the most preferrable worlds among the worlds that are doxastic-accessible. The idea of desires having a doxastic modal base and a preference flavor was developed by Heim (1992) and von Fintel (1999), in their Better-Worlds and Best-World Approach, respectively. The terms used to describe these two approaches are given by Grano (2021).

2.3.3.1 The Better-Worlds Approach

Heim (1992) describes the semantics of *want* as a combination of conditional semantics inspired by Stalnaker (1984), semantics for belief and semantics for

[5] Based on the semantics of belief reports as described by Hintikka (1969).

preference. At every world that is compatible with the subject's beliefs, Heim suggests that we check to see whether maximally similar worlds in which the subject teaches Tuesdays and Thursdays next semester are all more desirable to the subject than maximal similar worlds in which the subject is not teaching Tuesdays and Thursdays. If this is the case for each of the subject's belief worlds, then the sentence is true; otherwise, the sentence is false (Grano 2021: 157).

In this view, *want*-sentences seem to have the same semantic representation as *wish* sentences. This is not the case however, because *wish*-sentences are about worlds that are not among the attitude holder's doxastic alternatives, whereas *want*-sentences are propositions about what is possible to happen. To solve this, Heim proposed that the Sim function should select among worlds that are compatible with the subject's beliefs (21).

(21) The Better-Worlds Approach (Heim 1992):

$[[\text{I want to teach Tuesdays and Wednesdays next semester}]]^{W} = \forall w' \in DOX_{s,w}$:
$Sim_{w'}(DOX_{s,w} \cap \text{teaching Tuesdays and Wednesdays next semester}) >_{s,w}$
$Sim_{w'}(DOX_{s,w} \cap \neg \text{teaching Tuesdays and Wednesdays next semester})$[6]

(Grano 2021: 159)

According to the better-worlds approach, 'I want to want to teach Tuesdays and Wednesdays next semester means that in every world that the subject believes as possible, the worlds where she is teaching Tuesdays and Wednesdays that are similar to what she believes as an ideal world are more preferable than the worlds where she is not teaching Tuesdays and Wednesdays.

2.3.3.2 The Best-Worlds Approach

von Fintel (1999) approaches the semantics of *want* based on the semantics of modals, suggesting that the modal base is a combination of the attitude holder's beliefs and desires. The *best-worlds* are the worlds where both doxastic and bouletic alternatives are satisfied. The semantics of *want*, according to this approach is illustrated in (22).

(22) The Best-Worlds Approach (von Fintel 1999):

$[[\text{Beatrix wants it to be raining}]]^{W} = \forall w' \max_{BOUL_{b,w}}(DOX_{b,w})$: raining(w')

(Grano 2021: 160)

In (24), 'Beatrix wants it to be raining' means that in all the worlds that Beatrix considers as the most desirable and believes as possible, it is raining. If it is raining in all the worlds that are part of Beatrix's beliefs and best satisfy her desires, then the sentence is true. In the same way, the semantic analysis of *want* should describe that the desired situation holds in all the individual's most preferable possible worlds (23).

[6] Instead of Heim's (1992) 'less-than' (<) symbol for comparative desirability, Grano (2021: 156, footnote 2) uses the 'more than' (>) symbol as a more intuitive way to say 'more desirable'.

(23) $[[\text{want}]]^{w} = \lambda p.\lambda x.\ \forall w' \in \text{BEST}_{(w,x)}(D)\text{: } p(w') = 1$

In (23), it is suggested that *"an agent x wants p in world w iff all of the worlds in the relevant domain D that are "best" as far as x in w is concerned are p-worlds"* and *"the default value for D is the set of worlds compatible with x's beliefs in w"* (von Fintel and Iatridou 2023: 34-35).

2.3.3.3 The Antifactivity Approach: commitment in the Ideal worlds

Giannakidou and Mari (2021) propose that the predicate *want* is more similar to *must,* than to *believe.* They suggest that wanting is weaker than believing and argue that 'wanting p has an antifactivity presupposition meaning that either p is false at the time of utterance (24a) or the attitude holder believes p not to hold' (24b).

(24) a. Context: I just received a letter from Bill and I know it:
#I want so much to receive a letter from Bill!

b. Context: I just received a letter from Bill but I don't know it yet:
I want so much to receive a letter from Bill!

The semantics they propose for *want* is presented in (25). The bouletic state Boul expresses 'the will of doing' as *want* is defined as a set of worlds associated with an individual i representing worlds compatible with i's desires.

(25) $[\![\text{iWANTp}]\!]^{i,Dox,Boul,Ideal,S,tu}$ is defined iff $\forall w' \in Dox(i) : \neg p(w')$
(antifactivity), and Boul is nonveridical:
$\exists w'' \in Boul : \neg p(w'') \land \exists w''' \in Boul : p(w''')$
If defined,
$[\![\text{iWANTp}]\!]^{i,Dox,Boul,Ideal,S,tu} = 1 \text{ iff } \forall w'''' \in IdealS : p(w'''')$
(Giannakidou & Mari 2021; 204)

Want is defined as the bouletic equivalent of *must*. It does not express commitment in general, like hope, but only in the ideal worlds of the attitude holder. The antifactive *want* always selects for subjunctive. There are sentences where *want* is not antifactive. *Want* is sensitive to antifactivity if it appears in a context with temporal parameter, such as a finite sentence. If it appears in a context without temporal parameter, then it is a generic *want* (26).

(26) I want you to be free.

In this approach, wishes as X-marked desires that appear in environments with past tense morphology must be antifactive.

2.3.4 Differences between *want* and *believe*-sentences

The analysis of the meaning of *want*-sentences has been related to the analysis of beliefs. The semantic properties that point out the differences between *want* and believe-sentences are the incorporation of belief semantics to desire structures and vice versa, monotonicity, conjunction introduction, acceptance of conflicting desires or beliefs without leading to contradiction, gradability and focus sensitivity (Grano 2021).

Both the *Better* and the *Best-worlds* approaches predict that *a believes b* entails that *a wants b*. This means that *want*-sentences fail as presuppositions as they are entailed by *belief*-sentences. The sentence '*Beatrix believes that it is raining'* does not entail that 'Beatrix wants it to be raining'. This issue leads to the conclusion that *want*-sentences incorporate belief semantics, whereas *believe*-sentences do not incorporate desire semantics.

Another issue raised by the two approaches is that the *best-worlds* approach predicts monotonicity, whereas the *better-worlds* approach does not. Monotonicity predicts that *if a wants p* is true, and any proposition p entails q, then *a wants q*. If the *want*-sentence is monotonic, then it is upward-entailing on p which is the most desirable doxastic alternative of the speaker. The better-worlds approach does not predict upward-entailment. The analysis of *want* by von Fintel and Iatridou (2023; footnote 58) does not require widening of the domain to include non-p worlds and thus it does not predict monotonicity either: "Bringing non-p worlds into the comparison means that the "more upward" we go, the further out in the similarity ordering we may need to go to find non-p worlds and thus we may find a disruption of upward inferences. We do not currently see a way to combine monotonicity with a semantics that does not widen the domain to include non-p worlds".

Another semantic property that distinguishes beliefs from desires is conjunction introduction. In contrast to *belief*-sentences (27a), *want*-sentences do not obey conjunction introduction (27b).

(27) a. If a believes p and a believes q $\vDash$ a believes p $\wedge$ q.
b. If a wants p and a wants q $\nvDash$ a wants p $\wedge$ q.

(based on Grano 2021: 168)

This means that if *the attitude holder believes a* and *the attitude holder believes b*, it is necessary that the attitude holder believes a and b. However, if *the attitude holder wants a* and *the attitude holder wants b*, it is not necessary that *the attitude holder wants a and b*. For example, in a scenario where I am planning to go to the cinema to watch a film, I can express my desire to watch more than one films, which is possible, but also limit my desire and say that I want to watch only one film (28a).

(28) a. I want to watch 'Perfect Days' tonight and I want to watch 'Poor Things' tonight, but I don' t want to watch 'Perfect Days' and 'Poor Things' tonight.
b.*I believe I will watch 'Perfect Days' tonight and I believe I will watch 'Poor Things' tonight, but I do not believe I will watch 'Perfect Days' and 'Poor Things' tonight.

Belief-sentences do not allow for conflicting desires, because it leads to contradiction (29b), whereas *want*-sentences should (29a).

(29) [I want to watch 'Perfect days' and 'Poor Things' at the cinema. Tonight, I am going to the cinema and I am more in the mood for 'Perfect Days'. I decide to watch 'Perfect Days' tonight and 'Poor Things' next week. However, a lady at the entrance says that tonight is the last screening of 'Poor Things' at the cinema. The screening time for both films is 10pm, and I do not know which one

to choose and I say:]

a. I want to watch ‘Perfect Days’ at the cinema tonight at 10pm and I want to watch ‘Poor Things’ at the cinema tonight at 10pm.
b. *I believe I will watch ‘Perfect Days’ at the cinema tonight at 10pm and I believe I will watch ‘Poor Things’ at the cinema tonight at 10pm.

The example in (29a) shows that *want*-sentences allow for conflicting desires, without leading to contradiction. My desire to watch these two films at the cinema, presupposes that there are two such films being shown at the cinema. More particularly, the presupposition is that I believe that there is a cinema, where I can watch these two films that exist according to my beliefs. Desiring to watch both films, even though I know that it is not possible, is acceptable. Believing that I will watch both films simultaneously though, leads to contradiction. Beliefs do not rely on desire-semantics, but desires are dependent on beliefs, as they incorporate belief semantics.

Another property of *want* is that it behaves like a gradable predicate. The fact that preferences can be ranked in different ways shows that *want*-sentences are gradable (30a), whereas beliefs are not (30b).

(30) a. I want to watch ‘Perfect Days’ more than I want to watch ‘Poor Things.
b. *I believe I will watch ‘Perfect Days’ more than I believe I will watch ‘Poor Things.

Desires are also sensitive to focus, as the truth conditions of want-sentences depend on the focused complement of *want* (31).

(31) a. I want to COOK stuffed vine leaves (and not order them).
b. I want to cook STUFFED VINE LEAVES (and not beans).

The differences between *want* and *believe,* as summarized in Table 1, reflect their semantic properties.

Properties of verbs	**Want**	**Believe**
Incorporates desire or belief semantics	Yes (belief)	No
Monotonicity	Yes (von Fintel 1999) / No (Heim 1992, von Fintel Iatridou 2023)	No
Conjunction introduction	No	Yes
Accepts conflicting desires without leading to contradiction	Yes	No
Gradability	Yes	No
Focus-sensitivity	Yes	No

Table 1 *Semantic properties of want and believe (Grano 2021: 178)*

2.3.5 Towards the semantics of wish-sentences

The description of the semantic properties of *want* and *believe*-sentences is now applied to *wish*-sentences as a first step towards approaching the meaning of wishes. The incorporation of belief semantics can be a property of wish sentences. The speaker who wishes 'it were raining', Beatrix let's say, considers worlds that could be true (32). Beatrix's preference for other worlds is true even if in the actual world she knows that her desire is unattainable.

(32) Beatrix wishes it were raining.

Beatrix knows that in the actual world it is not raining. The fact that she wishes it were raining, shows that she has the belief that in an alternative world it is possible that it was raining. The subject expresses an unattainable desire in worlds that she considers as true alternatives to the actual world. In other words, in X-marked desires there is no limitation to the beliefs of the attitude holder (Wimmer 2024).

The next step is to see whether monotonicity is a semantic property of *wish*-sentences. Monotonicity is a property of some expressions to maintain or increase their truth value when the domain of the discourse changes to a smaller or larger context. Monotonicity is not a property of *want*-sentences because it cannot be combined with a semantic analysis that does not include domain widening. This might be an indicator that monotonicity is a semantic property of wishes as they require domain widening. Monotonicity in wishes is an issue to be explored.

Conjunction introduction is not a semantic property of *wish*-sentences, as shown in (33). Gradability (34) and focus (35), on the other hand, work perfectly with *wish*-sentences.

(33) *I wish I had watched 'Perfect Days' at the cinema last night at 10pm and I wish I had watched 'Poor Things' at the cinema last night at 10pm.

(34) I wish I had watched 'Perfect Days' more than I wish I had watched 'Poor Things'.

(35) a. I wish I had COOKED stuffed vine leaves (and not order them).
b. I wish I had cooked STUFFED VINE LEAVES (and not beans).

In Table 1, we saw the semantic properties of *want* and *believe*. In Table 2, I have added the verb *wish* to sum up the semantic properties of *wish*, *want* and *believe*.

Properties of verbs	**Wish**	**Want**	**Believe**
Incorporates desire or belief semantics	**Yes**	Yes (belief)	No
Monotonicity	**? (probably yes)**	Yes (von Fintel 1999) / No (Heim 1992, von Fintel Iatridou 2023)	No
Conjunction introduction	**No**	No	Yes
Accepts conflicting desires without leading to contradiction	**No**	Yes	No

Gradability	**Yes**	Yes	No
Focus-sensitivity	**Yes**	Yes	No

Table 2 *Semantic properties of wish*

The semantic properties that are found in *wish*-sentences are the use of belief semantics, gradability and focus sensitivity. These properties are also properties of *want*-sentences. *Wishes,* however, do not accept conflicting desires without leading to contradiction, whereas *want*-sentences do.

The *wish*-sentence 'Beatrix wishes it were raining' means that all the worlds that Beatrix desires the most, are not included in her doxastic set. This means that Beatrix knows that all the worlds where it is raining, which are the worlds that she desires the most, are counterfactual. Therefore, Beatrix knows that her desire is unattainable in the actual world.

Want-sentences express that the attitude holder desires something and knows that it is attainable in the actual world. However, the *want*-sentence 'Beatrix wants it to be raining' does not express a desire about the worlds that are among Beatrix's doxastic set. If all worlds where desires are fulfilled are among the attitude holder's doxastic set, then it is a *be-glad* sentence that expresses a guaranteed desire. In (36), the subject is happy because the desire to be raining is fulfilled.

(36) Xerome pu vreçi!
be.glad.1SG.NPST.IPFV that rain.3SG.NPST.IPFV
'I am glad it is raining.'

Want-sentences are similar to *wishes*, but they cannot express unattainable desires like wishes. '*Wish*-sentences are used to say something about the relative desirability of worlds that are not among the attitude holder's doxastic alternatives' (Grano 2021: 157). If *want*-sentences have a doxastic modal base with a preference ordering, then *wish*-sentences have a modal base that contains worlds that the subject believes to be counterfactual. In other words, *wish*-sentences have a wider domain than *want*-sentences. This suggests that sentences with X-marking have a wider domain than sentences with O-marking.

3 The components of X-marked desires in Modern Greek

3.1 The form of X-marked desires in Modern Greek

In Modern Greek, X-marked desires are formed as follows. Either there is X-marking morphology on the desire predicate, i.e. *tha ithela* (X-marked 'want') and on its complement, or instead of an X-marked verb, there is a wish dedicated word, with no X-marking, like the verb *efxome* ('I wish') or an optative marker like *makari* along with X-marking on the complement. X-marking morphology on the complement of the desire predicate in Modern Greek is expressed with pluperfect or past imperfective. The examples in (37) present these forms.

(37) a. Tha ithela na ixa fitepsi ena dendro.
MOD want.1SG.PST,IPFV NA-SUBJ plant.1SG,PLUP a tree
'I wish I had planted a tree.'

b. Efxome na ixa fitepsi ena dendro.
wish.1SG.NPST NA-SUBJ plant.1SG,PLUP a tree
'I wish I had planted a tree.'

c. Tha ithela na fiteva ena dendro (xtes).
MOD want.1SG.PST,IPFV NA-SUBJ plant.1SG,PST.IPFV a tree (yesterday)
'I wish I would have planted a tree yesterday.'

d. Efxome na fiteva ena dendro (xtes).
wish.1SG.NPST NA-SUBJ plant.1SG,PST.IPFV a tree (yesterday)
'I wish I would have planted a tree yesterday.'

The use of pluperfect in the complement of desire verbs and the consequent of counterfactual conditionals denotes counterfactuality. In (39a) and (39b) the complement of the desire predicate is in pluperfect, showing that the presence of pluperfect in such structures in Modern Greek denotes X-marking.

In (37c) and (37d) the complement of the desire predicate has past imperfective morphology which yields an ambiguity. It can either convey the meaning of X-marking as in (37c) and (37d), where there is the temporal adverbial *xtes* ('yesterday'), or it can have a Future Less Vivid meaning. The term Future Less Vivid (FLV) refers to the situation that is future oriented but is less possible to be fulfilled (Iatridou 2000). FLV desires are O-marked, as they refer to a desire that can be fulfilled in the near future though it is more impossible than possible that this will happen.

As regards the form of the desire predicate, in (37a) and (37c) there is X-marking morphology on the desire verb *thelo* ('I want'), whereas in (37b) and (37d) there is a wish dedicated word, like the verb efxome ('I wish') without any X-marking morphology on it. X-marking morphology of the desire predicate in Modern Greek, if any, is always past imperfective.

In general, X-marking on desires is ambiguous. It either expresses an endo-X desire, that is an unattainable desire in the actual world (38a) or an exo-X desire, that is an attainable desire in a counterfactual scenario (38b).

(38) a. Tha ithele na ixe megalitero krevati.
FUT want.3SG.PST.IMP NA have3SG.PST bigger bed.
'He wishes he had a longer bed.'

b. An itan psiloteros, tha ithele na exi megalitero
if be.3SG.PST taller, FUT want. 3SG.PST.IMP na have3SG,NPST.IMP bigger
krevati.
bed.ACC
'If he was taller, he would want to have a bigger bed.'
(von Fintel and Iatridou 2023: 1495)

In Greek, endo-X and exo-X desires appear with the same form. In both (44a) and (38b), X-marking appears on the verb *thelo* ('want') through past imperfective morphology. In (38b) the exo-X desire is formed though X-marking applied on the verb *thelo* and an O-marked complement, which is a verb in present tense. In (39a, b) both desires are exo-X. However, they differ in the complement of the X-marked *thelo*, as in (39a) it is in non-past tense, whereas in (39b) the complement of *tha ithela* is in past tense. This means that there can be an unattainable desire in a counterfactual scenario.

(39) a. An o Haris itan psiloteros, tha ithele na agorasi
if the Haris be.1SG.PST taller, FUT want.3SG.PST.IPFV NA buy.3SG.NPST.PERF
megalitero krevati.
bigger bed.
'If Haris was taller, he would want to buy a bigger bed.'

b. An o Haris itan psiloteros tha ithele na itan ke
if the Haris be.1SG.PST taller FUT want.3SG.PST.IPFV NA be.1SG.PST and
Amerikanos.
American.
'If Haris was taller, he would wish that he was also American.'
(...so that he could be drafted to the NBA more easily)
(von Fintel and Iatridou 2023: 1494, footnote 45)

In (45a), it is suggested that there is one layer of X-marking in the desire proposition, as it is the consequent of an X-marked conditional. In (39b) there are two layers of X-marking in the desire proposition, one for being the consequent of an X-marked conditional and one for the unattainable desire. The desire propositions in (39a) and (39b) have X-marking on *thelo* and their complement is also X-marked. They differ however, in that (39a) is an X-marked desire in the world where the attitude holder lives and believes it to be true. In (39b) the X-marked desire is the consequent of an X-marked conditional. This means that the desire is unattainable in a counterfactual world.

In Modern Greek there is also the verb epithimo ('I desire') and other wish dedicated phrases like *ax ce* and *tulaxiston* that are used in X-marked desire constructions. They are presented in combination with the morphology of the wish complements in the following section that is about the subcategorization information of *want* and *wish* sentences in Modern Greek.

3.2 Subcategorization Information

The attitude verbs that are more commonly used to express desire and wish in Modern Greek are *thelo ('I want'), efxome ('I wish')* and *epithimo ('I desire')*[7]. Other verbs that also express desire in Modern Greek, but are not analyzed in this section, are *protimo* ('I prefer'), *onirevome* ('I dream'), *fandasionome* ('I fantasize'), *tha xeromun na* ('I would be glad to') and *tha imun xarumeni an* ('I would be happy if'). The optative element *makari,* as well as the phrases *toulaxiston* and *ax ce* introduce optative utterances that express wishes.

Wishes, in the sense of X-marked desires, are not formulaic. Formulaic wishes are DPs that express the speaker's desires or aspirations in occasions like holidays, birthdays, anniversaries, and celebrations in general. Formulaic wishes are formed either with bare exclamative DPs (40a) or with the verb *efxome* ('I wish') with a DP complement (40b). *Thelo, epithimo* and optative elements like *makari* do not appear with such formulaic wish DPs (40c).

(40) a. Kala xristuʝena!
good Christmas
'Merry Christmas!'

b. Efxome kala xristuʝena.
wish.1SG,NPST good Christmas
'I wish you Merry Christmas.'

c. * Thelo/ *epithimo/ *makari/ *toulaxiston/ *ax ce kala xristuʝena.
*Want/desire/hope.1SG,NPST/ MAKARI/ 'AT-LEAST'/ AX INTERJT good christmas
'I want/ desire/ if only/ if at least/ if only Merry Christmas.'

The desire verbs that do not accept a formulaic wish DP[8] can take a non-formulaic wish DP as a complement (41a). The verb *efxome,* does not accept a non-formulaic wish DP (41b).

(41) a. Thelo nero.
want.1SG.NPST water
'I want water.'

b. *Efxome nero
wish.1SG.NPST water
'I wish water.'

When the desire predicate *thelo* ('I want') is X-marked, it appears with past tense and imperfective aspect, and it is preceded by the modal particle *tha* ('would'). *Tha* is an

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[7] The list of verbs (and the tenses of verbs) presented in 2.4.1. is not exhaustive, *i.e. protimo ('I prefer'), onirevome ('I dream'), fandasionome ('I fantasize'), tha xeromun na ('I would be glad to'), tha imun xarumeni an ('I would be happy if'), tha efxomun ('I would wish'), tha epithimusa ('I would desire')*.

[8] I use the term formulaic wish DP, to refer to DPs like in (40), *i.e. kala xristugena ('Merry Christmas '), xronia pola ('Happy Birthday'),* and to the DPs that can be used in a wish-context, sometimes preceded by an adjective *(like kalos 'good'), i.e. kali dinami* ('good strength', meaning 'stay strong'), *kali oreksi* ('good appetite', meaning 'enjoy your meal' *), igia* ('health'), *kuragio* ('courage').

element that originally denotes futurity, but in conditional and desire structures its combination with CPs with past tense morphology shows that it expresses a conditional situation. *Tha* is part of the consequent of an X-marked conditional 'if $p_{ANTECEDENT}$ then $q_{CONSEQUENT}$' and it appears on the desire predicate of an X-marked desire 'I $want_{CONSEQUENT}$ that $p_{ANTECEDENT}$'.

The verb *efxome* can take a CP complement in past tense without having past tense morphology itself (42b). The desire predicates of X-marked desires that do not have a wish lexical meaning appear as X-marked. In (42c) the desire is X-marked, as indicated by the pluperfect morphology of the complement. The desire predicates *thelo* and *epithimo* are not X-marked, neither they have a wish-meaning. This is why the sentence in (42c) appears as ungrammatical. It is important to note that some native speakers of Greek accept *thelo* in (42c).[9] This shows that X-marking depends more on the morphology of the complement than on the desire predicate.

(42) a. Thelo na fitepso ena dendro.
want.1SG.NPST,IPFV NA-SUBJ plant.1SG,NPST,PERF a tree
'I want to plant a tree.'

b. Efxome/ tha ithela/ tha epithimusa/ makari na ixa fitepsi ena dendro
wish.1SG.NPST/ MOD want/ desire.1SG.PST,IPFV/ MAKARI NA-SUBJ plant.1SG,PLUP a tree
'I wish I had planted a tree.'

c. *Thelo/ *epithimo na ixa fitepsi ena dendro.
want.1SG.NPST/ desire.1SG.PST,IPFV NA-SUBJ plant.1SG,PLUP a tree
*'I want/ desire I had planted a tree in the yard.'

Furthermore, the optative and conditional markers that introduce optative utterances receive a wish lexical meaning. *Makari, ax ce* and *toulaxiston* introduce a main subjunctive clause and they behave like *efxome* and *tha ithela*. Their complement needs to be a CP, as it cannot co-occur with DPs (43).

(43) *Makari/ *toulaxiston/ *ax ce kala xristuʝena.
Makari/ 'at least '/ if- good christmas

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[9] Even though I do not accept (42c) as grammatical, I can imagine a scenario where *want* in present tense with an X-marked complement does not lead to ungrammaticality. The scenario is the following: A boy named Haris is crying because he is going to school with regular clothes, and he wanted to wear his Batman costume instead. When the teacher saw him crying loudly while getting out of his parent's car, the following dialogue occurred:

(i) Teacher: ʝati klei toso poli o xaris?
Why cry.NPST.3SG that much DET haris
'Why is Haris crying so much?'

Parent: theli na ixe dithi batman
want.NPST.3SG NA-SUBJ dress-up-PLUP batman
'He wishes he had dressed up as batman'

I think that in (i) the speaker uses *want* in present tense to associate the boy's desire (which is a desire that he still has) with the fact that he is crying at the moment of utterance. My idea is that a desire predicate in present tense is closer to expressing the feeling that the attitude holder has in the 'here and now', than a desire predicate in past tense. *Theli* is more expressive than *tha ithele* and the justification of an emotion maybe requires a more expressive form of the verb. This is an idea that needs further exploration and work.

'I wish you merry Christmas.'

If the CP complement does not have past tense morphology, the desire is O-marked (42a), and if it has it is X-marked (43a) or FLV. X-marked desires are formed with *efxome, tha ithela* or *makari* and a past tense CP complement.

The subcategorization information of these desire predicates in Modern Greek is summarized in Table 3.

Desire predicate	**Complement of the desire predicate (O-marking)**	**Complement of the desire predicate (X-marking)**
Thelo	_ Non formulaic-wish DP, _[- past] CP	-
Epithimo	_ Non formulaic-wish DP, _[- past] CP	-
Tha ithela	_ Non formulaic-wish DP, _[- past CP],	**_[+ past] CP**
Efxome	_Formulaic-wish DP, _[- past] CP,	**_[+ past] CP**
Makari	_[- past] CP,	**_[+ past] CP**

Table 3 *Subcategorization information of desire and wish words*

Table 3 shows that X-marked desires need to have a complement with past-tense morphology. If there is no past tense in the complement, and not necessarily on the desire predicate, then there is no X-marking in Modern Greek.

3.3 Past Tense

The use of past tense morphology, through pluperfect or past imperfective, in the complement of the desire predicate is necessary in X-marked structures in Modern Greek. When the desire predicate is X-marked it is always past imperfective, showing the necessity of past tense morphology.

Past tense in the time axis is used to show that the topic time comes before the utterance time (Klein 1994). A single past tense morpheme is used for a present X-marked desire and a double past morpheme is used for a past X-marked desire, showing that past is not necessarily interpreted as past in the time frame. FLV desires have the same morphology with present X-marked desires, and they are future oriented. The fact that a past tense morpheme is used in a sentence with a future reading indicates that past is not interpreted as past in X-marked and FLV structures. In these environments, past tense is used to denote counterfactuality. For this reason, it has been considered as 'fake'. Of course, past cannot be 'fake', but it has been characterized considered as such in comparison to its default function as past in the time frame. In this view, past tense in X-marked desires is 'fake'. X-marked desires with past imperfective have one layer of 'fake' past and X-marked desires with pluperfect have one layer of 'fake' past and one layer of real past.

There are two approaches on how tense is interpreted in X-marking environments. There is the past as modal and the past as past approach (Schultz 2014), alternatively referred to as the fake and the real past approach (Karawani 2014).

According to the theory of past as modal, the past tense morpheme is either fed times, and it is evaluated in a time that is before the utterance time in the time axis, or it is fed worlds and it is evaluated in a world that is different from the actual world (Iatridou 2000, Schultz 2014, Mackay 2019). The past tense morpheme denotes an exclusion feature, and it departs either from the moment of utterance (44a) or from the actual world (44b).

(44) Past tense as an exclusion feature:
 a. Time: T(t) excludes C(t) = The topic time excludes the utterance time
 b. Worlds: T(w) excludes C(w) = The topic world excludes the actual world

(Iatridou 2000: 246-7)

In this view, the use of past as modal over worlds leads to the modal effect of X-marking, as depicted in Figure 1.

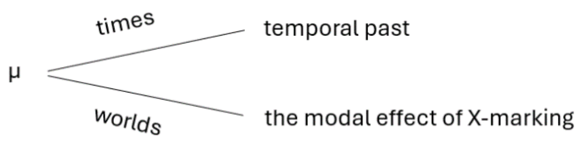

Figure 1 Past as modal (von Fintel & Iatridou 2023:1478)

According to the theory of past as past, the time of the evaluation of the past tense morphemes in counterfactuals is shifted in the past and it is interpreted epistemically (Ippolito 2003, Arregui 2005, Khoo 2015). There is a contradiction in the unified theory of past as real, as it treats past as being the same in all environments and changing after a splitting point in the time axis, showing the alternatives to the actual world, as depicted in Figure 2. This hypothesis relies on the inherent nature of past being closed and thus expressing impossibility for fulfillment. It does not explain, however, how past tense morphology contributes to the meaning of a past evaluation point out of counterfactual structures (Karawani 2014).

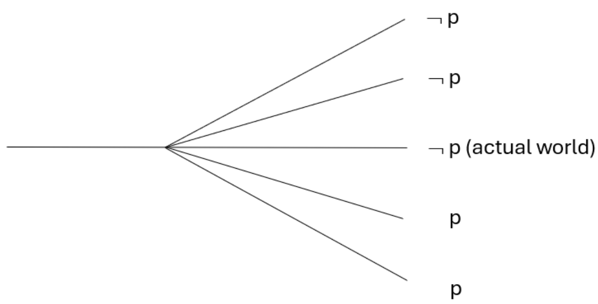

Figure 2 Past as past (von Fintel & Iatridou 2023:1479)

The past as past view treats X-marking as being epistemic. In the past as modal view, the 'fake' past that was later described as a modal that scopes over worlds, was proposed to exclude the actual world and therefore the X-marking modal base is not at all contained in the epistemic set. In X-marking, however, the modal base is not entirely contained in worlds that are epistemically accessible. The epistemic set contains all the situations that are O-marked, but not all the situations that are X-marked. The past as past view suggests that X-marked structures are entirely contained in the epistemic set. The past as modal theory on the other hand, suggests that the domain of quantification in X-marked structures is entirely excluded from the epistemic set. A solution to this problem would be an explanation of past tense in X-marked structures that considers the need for domain widening.

3.4 Imperfective Aspect

X-marked desires with past imperfective morphology receive a perfective interpretation, exactly as it happens with conditionals with past imperfective morphology. In (45a), imperfective aspectual morphology is used, even though the meaning of the sentence is not that *if the subject will be in the process of taking that syrup, they will be in the process of getting well*. When perfective aspect is used in conditionals, it can only be interpreted epistemically and the conditional is not X-marked. When past tense is used with perfective aspect, the sentence is O-marked (45b).

(45) a. An eperne afto to siropi tha ginotan kala.
if take.PST.IPFV this the syrup FUT become.PST.IPFV well
'If he took this syrup, he would get better.'

b. An ipxe afto to siropi (tha/ prepi na) egine kala.
if drink.PST.PRF this the syrup (MOD/ must NA) become.PST.PRF well
'If he drank this syrup, he must be better.'

(Iatridou 2000: 236-7)

In contrast with past tense that can appear in two layers, as the double past tense marking in pluperfect, imperfective aspect appears through one morpheme. X-marked desires and conditionals appear only with imperfective aspect, as there is no room for perfective aspect and so we cannot talk about a real and fake aspect distinction. In Modern Greek, imperfective aspectual morphology 'comes for free' with past tense morphology (Karawani 2014). X-marked desires must contain imperfective aspect (46a), otherwise the desire is not X-marked (46b). The fact than adverbs that express completion, like 'in one day' in (46) and are used in perfective environments can also be used in FLV environments, where imperfective aspect is used, shows that the sentence is interpreted perfectively (Iatridou 2000: 236).

(46) a. Tha ithela na ginotan kala (se mia mera).
MOD want.1SG.PST.PRF NA become.1SG.PST.IPFV well (in one day)
'I wish he would get well (in one day).'

b. Tha ithela na egine kala (se mia mera).
MOD want.1SG.PST.PRF NA become.3SG.PST.PRF well (in one day)
'I wish he got well (in one day).'

Imperfective aspect is a necessary component of the X-marking morphology of the desire predicate in X-marked desires. However, it is not needed in its complement. An alternative way to express X-marked desires language, that is slightly different from what we have discussed so far, is through optative utterances. The role of optativity is being described in the following section.

3.5 The role of optativity

Optative constructions can be realized with optative particles like *makari.* Even though there is no morphological expression of optative mood in Modern Greek, optativity is present through the combination of optative particles and optative utterances that are exclamations that show the speaker's emotions.

Kyriakaki (2009) suggests that wishes in Modern Greek contain a covert operator that is exclamative in nature and has a wish reading. She proposes that, what is here described as a main subjunctive wish, is in fact an exclamatory wish that allows only for exclamatory wish particles like *makari,* and interjections like *ax ce* (47).

(47) Makari/ ax ce na min evrexe toso!
EXCL.wish/ INTERJ NA NEG rain.3SG.PST.IPFV so-much.
'If only it didn't rain so much!'
(Kyriakaki 2009:12)

Optative constructions are exclamative utterances that are used to express wish, desire, regret and hope without words that have such related lexical meaning (Grosz 2012). The crosslinguistic pattern of optatives is that particles that mean 'only', 'but' and 'at least' can turn a conditional antecedent like 'If John had listened to Mary' in (48a) into an if-optative (48b) that can be paraphrased to an X-marked desire (48c).

(48) a. An o John ixe akusi tin Mary, tha ixe pari mia obrela.
if the John listen.3SG.PLUP the Mary MOD take.3SG.PLUP an umbrella
'If John had listened to Mary, he would have taken an umbrella.'

b. An tulaxiston o John ixe akusi tin Mary! (Grosz 2012:293)
if at-least the John listen.3SG.PLUP the Mary
'If John had at least listened to Mary!'

c. Tha ithela o John na ixe akusi tin Mary!
MOD want.3SG.PST.IMPERF the John NA listen.3SG.PLUP theMary
'I wish John had listened to Mary!'

Grosz (2012) proposes that optative utterances contain a scalar and expressive EX operator that takes the denoted proposition as its complement, and it expresses an emotion towards the denoted proposition. EX is scalar because it operates on scales as it reflects the speaker's preferences. An utterance like (48b) has an EX-operator, as it expresses the emotion of the speaker and thus it can be felicitous or infelicitous. An utterance like (48c) lacks the EX-operator, because it describes the desire of the speaker, but it does not show the expression of any emotion and thus it can be true or false.

Tulaxiston shows the speaker's desperation in the actual situation, but also their willingness to compromise. This is the same in languages, where 'at least' appears in optative utterances. In (48b), *tulaxiston* serves as the EX-operator that takes the proposition 'John had listened to Mary' as its complement and it expresses the emotion of the speaker towards the proposition (49).

(49) [[EX Scale$_{\text{speaker-preference}}$] [John had at least listened to Mary]] ≈ The desirability of [p John had at least listened to Mary] exceeds a contextually salient threshold ξ.

In (51b), as shown in the optative reading in (49), the speaker expresses an emotion towards the unlikelihood that 'John listened to Mary' on the speaker's desirability scale. The threshold ξ corresponds to "*the cut-off line between what is not surprising (lower/more likely) and what is surprising (higher/more unlikely)"* (Grosz 2012: 41). In (48c) 'I wish John had listened to Mary' is true if and only if the counterfactual proposition 'John had listened to Mary' is ranked above a contextually salient threshold on the speaker's desirability scale.

Human behavior has a cognitive and an emotional dimension that is reflected in grammar and more specifically in the difference between (50b) and (50c). In formal terms, (50b) has an EX-operator, that (50c) lacks. In (50b) the speaker expresses an emotion towards the unlikehood of fulfilling what is more desirable. In (50c) the speaker describes what is more desirable but does not show any emotion.

The EX-operator shows that the utterance is exclamative and expressive. The lack of it results in absence of exclamation and expression. In (54a) the EX-utterance reflects the speaker's surprise, whereas in (54b) the utterance that lacks the operator EX, indicates the speaker's description of the feeling of surprise.

(50) a. Boy, is this easy!
b. I am surprised at how easy this is!

(Grosz 2012:45)

The generalization that occurs is that utterances without EX are linked to the cognitive dimension of human behavior and utterances with EX reflect its emotional dimension, as shown in Figure 3.

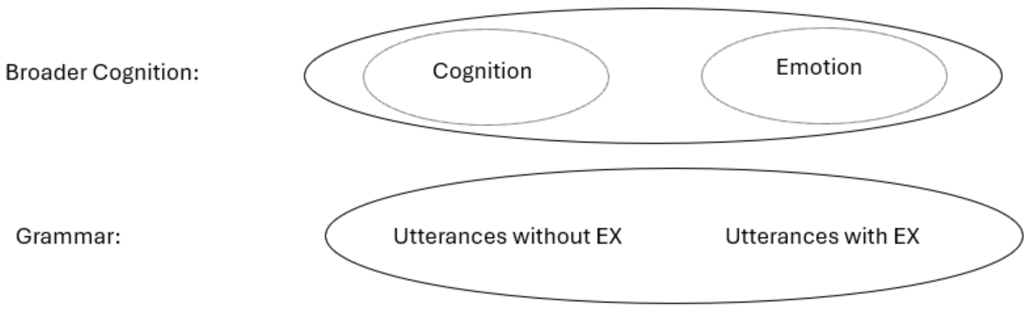

Figure 3 The cognition – emotion dichotomy (Grosz 2012:44)

Optative utterances are exclamative utterances in nature. They show that the speaker expresses emotions towards the proposition that is preferable and thus desirable. The hypothesis that wishes contain an operator that shows their exclamatory

(Kyriakaki 2009) and expressive nature (Grosz 2012) contributes to understanding the different types of wishes, i.e. expressive versus descriptive, and to a further extent their reflection in human behavior.

4 Previous studies on the development of wishes

To the best of my knowledge, the development of unattainable desires has not been studied experimentally. Evidence regarding the development of wishes comes from naturalistic longitudinal data (CHILDES database) for English (Tulling & Cournane 2019, 2022).

4.1 Tulling & Cournane (2019, 2022)

Tulling and Cournane (2019, 2022) studied the role of “fake” past tense in counterfactual wishes and conditionals through a corpus study, with data from CHILDES, on the spontaneous production of English-speaking children between the ages of 2 to 6. They set two questions. First, they wondered whether children produce *wish*-sentences that lack the “fake” past tense, and they predicted that children use past tense to express desires about the past and present tense to express desires about the present. The second question they made was whether counterfactual wishes are produced before or after counterfactual conditionals. They predicted that children produce counterfactual wishes before counterfactual conditionals, as the form-to-meaning mapping of wishes is easier to acquire than that of conditionals.

The authors suggested that the acquisition of counterfactual constructions comes after the development of simple future conditionals because of the cognitive load of counterfactual reasoning that demands to hold many possibilities in mind and to consider a false possibility as temporarily true. The fact that past tense morphology can either refer to the actual past or to a “fake” past introducing counterfactuality led to the first hypothesis. Acquiring counterfactual wishes demands that children understand that the verb ‘I wish’ cannot take a present tense complement and that wish-desires are different from want-desires. This mismatch between the temporal orientation and the tense marking of the complement of ‘I wish’ can help children understand the “fakeness” of counterfactual past tense, facilitating the development of counterfactual conditionals. This thought led the second prediction.

The methodology they used was a corpus-based analysis of data extracted from CHILDES, using the database ‘childes-db’ (Sanchez et al., 2019). They used the statistical software environment R to access the database. They collected data from typically developing monolingual children between the ages of 2;5-6;0 years old. After excluding wishes as nouns and wishes where the complement is not a full proposition (NP, PP complement, absence of complements), they kept data from 6 children. A further criterion was that children produced more than 15 wishes.

In their first study, Tulling and Cournane (2019) extracted 398 counterfactual wishes and 381 conditionals. They excluded the conditionals that were repetitions of the parent’s utterance and did not have a counterfactual use and the wishes that did not have a full proposition complement. In their 2022 study, they found 349 wish-constructions. The first wish in the corpus was uttered at the age 2;1 (51a) and the first counterfactual wish with a past temporal orientation was found at the age of 3 (51b). The first counterfactual conditional with past temporal orientation was found at the age of 3;8 (51c).

(51) a. Laura (2;1): I wish I had sandals.

b. Thomas (3): Your wish you gotten on this train.
c. Abe (3;8): no he would have smelled really bad if he died.
d. Adam (5;2): I wish I have a banjo like dis.
e. Sarah (3;6): I wish it's valentine.
f. Abe (4;4): I wish I asked for toast instead.

The results showed that the most frequent error (13% in Study 1(2019) and 10.9% in Study 2 (2022)) was the use of present instead of past tense in the wish complement (51d, e). Children were also found to express wishes about the past without using the past perfect (51f). Children's first wishes are not adultlike, and they are used like simple direct desires. The results of the second study showed that children use the verb 'I wish' like the verb 'I want'. Children's errors show that they are in the process of acquiring the interpretation of past tense as counterfactual in wish constructions, facilitating the acquisition of the less transparent counterfactual conditionals.

The overall results confirmed both that children make productive present-for-past errors from age 2 – 4, and that wishes emerge before counterfactual conditionals. It was pointed out though, that 2 out of 6 children, did not follow this developmental path. The authors suggested that counterfactual constructions are difficult to acquire as they require counterfactual reasoning and a complex form-to-meaning mapping.

4.2 Questions raised by wishes for the acquisitional studies

The linguistic expression of desires and wishes raises the following questions for the acquisition studies:

1. When do children acquire past tense with its modal use?
2. When and in what stages does irrealis mood develop in comparison to realis?
3. How does counterfactual reasoning develop?
4. What can the development of counterfactual emotions show for the development of counterfactual reasoning and counterfactuality in language?

These questions aim to answer the broader question about the development of counterfactuality, that is essential to acquiring unattainable desires. The acquisition of the 'fake' past tense that denotes counterfactuality and the development of irrealis modality, counterfactual reasoning and counterfactual emotions are key to exploring the developmental path of wishes. (Crutchley 2004; de Villiers 2005; Beck et al 2006; Rafetseder, Christi-Vargas & Perner 2010; Rafetseder, Schwitalla & Perner 2013; Rafetseder & Perner 2010, 2011, 2014; Beck & Riggs 2014; Rouvoli, Tsakali & Kazanina 2019; Tulling & Cournane 2019, 2022; Kazanina, Baker and Seddon 2020; Gómez-Sánchez, Ruiz-Ballesteros & Moreno-Ríos 2020).

4.2.1 Acquiring the modal past

The development of counterfactual wishes and conditionals requires the attribution of counterfactual meaning to the "fake" past tense morpheme, at least in transparent wish languages like Greek. The past tense morpheme can be used either as ‘real’ to express past in the time frame or as ‘fake’ to express counterfactuality (Iatridou 2000). This second meaning of the past tense morpheme is less common than its first use and therefore it is more challenging to be acquired by children. Children make errors while using the ‘fake’ past tense that denotes counterfactuality and they use present instead of past to denote non-actuality in the present (Tulling & Cournane 2019, 2022).

The reason why younger [10] children fail with past counterfactuals is not a difficulty with counterfactual thinking, but a lack of the linguistic awareness that *if + pluperfect* denotes counterfactuality (Rouvoli, Tsakali & Kazanina 2019). Mastering counterfactuality presupposes understanding conditionals, representing counterfactual worlds, and knowing that *if + past perfect* denotes counterfactuality. Rouvoli, Tsakali and Kazanina (2019) suggest that children’s failure with past counterfactuals stems from their lack of understanding the linguistic form and not from their inability to think counterfactually. For example, younger children cannot understand the hidden negation of (52), which is that *Amy did not win the medal*.

(52) If Amy had eaten the apple, she would have won the medal.

To test their hypothesis that the linguistic understanding of the grammatical construction *if + present perfect* is additional to the child’s counterfactual reasoning abilities, they conducted a past counterfactual experiment. 74 children, from 3;1 to 6;11 years old, watched video stories about a food contest where different animals were awarded either a medal or a cross for choosing to eat healthy or not-so-healthy foods. The experimenter held a puppet that watched every video with the participants and made the question at the end. Apart from the act out task that tested past counterfactuals, the researchers also designed a false belief task, to test children’s ability to entertain possible worlds. A future conditionals task was used to familiarize participants with the rules of the game and to evaluate their comprehension of future conditionals. The main task targeted at showing whether children understand that *if + past perfect* constructions are counterfactual with sentences like (53), where Mr. Penguin ate a snack and received a cross and a different animal ate a healthy food and received a medal in the actual world, as represented by the video story. The child is asked to guess the animal the puppet is referring to.

(53) If he had eaten a watermelon, he would have got the medal.

The correct answer in (53) is “Mr. Penguin” and it requires children to understand that Mr. Penguin did not eat a watermelon or get a medal in the actual world. The results showed that even though children answered correctly to the false belief questions (correct answers per age group: 3-year-olds: 87%, 4-yo: 100%, 5-yo: 76% and 6-yo: 100%), they made errors on the past counterfactual task (correct answers per age group: 3-year-olds: 23%, 4-yo: 86%, 5-yo: 92% and 6-yo: 99%). The authors interpreted these errors as a lack of linguistic knowledge that *if + pluperfect* denotes counterfactuality.

Crutchley (2004) explored older children’s production of past counterfactual conditionals and showed that there is systematicity in the errors children make while

[10] Children younger than 6 years old are characterized as ‘younger’ in this chapter.

acquiring conditionals. Through an elicitation task, she offered new data on the emerging control of past counterfactual structures with a sample of 799 typically developing children. She used the ACE 6-11 (Assessment for Comprehension and Expansion) tool, which was developed by Adams et al. (2001), that focuses on the development of the syntactic structures acquired early, like simple past tense, and later, like counterfactuals.

During the experimental procedure, every child was presented with a picture, showing an undesirable event, and a thought bubble of what should have been done to prevent the undesirable event. For example, in a scenario where a teacher shouts at her student of not doing her homework, the student thinks, as depicted in the thought-bubble, that if she had done her homework, she would not have been criticized. After pointing to the thought-bubble and then to the picture and explaining what they both show, the experimenter uttered the counterfactual conditional sentence and asked the child to repeat it. The results confirmed the main predictions of the study according to which children's production gets better with age and that all the errors or mismatched answers decrease. Younger children produce simpler conditionals and not the target structure and might describe the event rather than use the target sentence or even reply that they do not know. For example, in the teacher-student scenario, most younger children did not produce the adult-like counterfactual sentence 'If she had done her homework, she would not have been told off by the teacher', but they used past simple instead of past perfect. It was concluded that children combine *if* and *would-sentences* with verb tenses in a systematic way. This systematicity shows that children know how to form a conditional before speaking in an adult like manner (Crutchley 2004).

Younger children use present instead of past to denote non-actuality in the moment of utterance as they have not acquired the 'fake' past tense that denotes counterfactuality (Rouvoli, Tsakali and Kazanina 2019; Tulling and Cournane 2019, 2022). The development of such structures gets better with age (Crutchley 2004). Children's understanding of wishes requires the acquisition of the linguistic structures that denote counterfactuality.

4.2.2 Realis is acquired before irrealis

Younger children treat counterfactual utterances as realis and not irrealis and they might even deny giving a response (Reilly 1982). There is a general agreement in the literature that the interpretations of sentences that are not in conflict with reality are acquired first (Reilly 1982; de Villiers 2005; Beck et al 2006; Weisberg and Beck 2010; Rafetseder, Christi-Vargas and Perner 2010; Rafetseder and Perner 2010; Rafetseder, Schwitalla & Perner 2013; Kazanina, Baker and Seddon 2020; Gómez-Sánchez et al 2020).

The fact that younger children are not aware that past counterfactuals denote non-actuality is supported by the Actuality bias hypothesis in language acquisition, according to which that children first learn the syntactic and semantic constructions that are related to reality (de Villiers 2005 ΄ Kazanina, Baker and Seddon 2020).

Kazanina, Baker and Seddon (2020) suggested that young children's understanding of the sublexically modal verbs 'throw' and 'send' is not adultlike, whereas this is not the case with the modal verb 'want'. Given that want is understood by children already by the age of 2, according to Bartsch and Wellman (1995), the authors predicted that children would perform adultlike on sentences with 'want'. They conducted two experiments to explore whether younger children know that *X threw Y to Z* lacks a successful transfer entailment and that X wanted to throw Y to Z indicates that X had a desire or intention for Z to get Y, without entailing its fulfillment in the actual world.

In both experiments, the experimenter acted out a story based on the verbs ‘throw’ and ‘send’. In one of the stories, for example, two siblings, Jane and Tom, along with their pet sheep Woolly, went to play at the park. Tom threw a ball to Woolly, and it caught it, whereas when Jane threw a frisbee to Woolly, the wind blew, and it fell into Tom’s hands. In experiment 1, 59 3-4-year-old children watched such stories and listened to the story description by a puppet. Afterwards, they were asked to judge a simple ‘throw/ send’ sentence (54), mentioning either the intended recipient (54a) or the actual recipient (54b), and answer a ‘want-to’ question (55).

(54) a. Jane threw the Frisbee to Woolly.
b. Jane threw the Frisbee to Tom.

(55) Who did Jane want to throw the frisbee to?

The results suggested that children rejected (54b), with an acceptance rate of 49%, unlike adults who accepted 90% of such sentences, showing that they have a strong bias towards reality and that their performance on want-to sentences was like that of adults (86% correct responses). In experiment 2, the animate actual recipient was substituted by an inanimate one (i.e. trees) to check whether a more salient actual recipient, in comparison to the intended recipient, affects children’s answers. 120 3-6-year-old children were tested, and the results were like those of the first experiment. Children gave actual responses (60% acceptance) in sentences like (58b) and had an even better performance rate on want-to sentences (92% correct responses). Overall, children had difficulty interpreting verbs ‘send’ and ‘throw’ that encode modality, as they gave actuality responses regardless of the animacy of the recipient. Developing de Villiers’ s (2005) idea on reality bias, the authors suggested that there is an Actuality bias on verb learning.

Jill de Villiers (2005) discussed the acquisition of complement-taking verbs with regards to Theory of Mind. In her analysis of the verb ‘want’, she claimed that verbs of desire take complements that have the property that their tense is ‘forward’ from the time of the desire, captured in English by a to-infinitive (56).

(56) He wanted to go to the party.

The author linked the emergence of the verb ‘want’ to children’s first connection to their own desires, as they yearn towards the object of their desire, but also to hearing it from others. Wanting an object, a ball for example, is different than kicking or holding it, in that it extends to the future with potential fulfillment in the actual world. The objects of desire are irrealis in that sense. Children start interpreting modal verbs based on positive evidence, that is events of the actual world, until they start using the verb ‘want’ with irrealis complements.

The difference between realis and irrealis events is that they distinguish between what is observable and unobservable at the moment of utterance, respectively (Bickerton 1981). Even until the age of 13, children have difficulty to distinguish between real and conjectured information. The other factors that contribute to this difficulty are inferencing and understanding the linguistic expressions that are dedicated to expressing counterfactuality (Gómez-Sánchez, Ruiz-Ballesteros & Moreno-Ríos 2020).

Beck et al (2006) explored children's understanding of future hypothetical conditionals and counterfactuals as possibilities. They suggested that children understand future hypotheticals correctly independently from their ability to entertain alternative worlds, but they are cognitively mature to master counterfactuals only after preschool years, when they have developed the ability to think on possible worlds.

They conducted two comprehension experiments to find whether children acknowledge both possible outcomes of an event yet unknown (undetermined trials), a future hypothetical (experiment 1) or a counterfactual (experiment 2) alternative. Children were divided in two groups, a group of 3-4 and a group of 4-5-year-olds. In both experiments children played a game in which a toy mouse would go down slide 1 or slide 2 and they had to answer to the question of the experimenter by using a mat under the slide they predicted the mouse would run down to as an alternative. The questions used in experiments 1 and 2 were ''What if next time he goes the other way?'' for a single hypothetical future event and 'What if he had gone the other way?'' for a single counterfactual event, respectively. An open counterfactual question, like "Could he have gone anywhere else?'', was also used in the second experiment to test children's ability to think of counterfactuals as alternative possibilities. It was predicted that children would succeed on the future hypothetical and counterfactual trials only if they could answer correctly on the undetermined trials, meaning that they have acquired the ability to think of multiple possible outcomes.

In contrast with that prediction, the results of both experiments showed that children of both age groups found the undetermined trials difficult (mean score of group1: 0,6/2 and group 2:1.2/2 for experiment 1, group1: 1/2 and group 2: 1.7/2 for experiment 2), whereas they answered correctly to future hypothetical (mean score of group1: 1.9/2 and group 2:1.8/2) and standard counterfactual (mean score of group1: 1.7/2 and group 2:2/2) questions. Children found open counterfactuals more difficult to understand (mean score of group1: 1.3/2 and group 2:1.7/2) but performed better than in the undetermined questions. The findings of the experiments led to the conclusion that children can think hypothetically, without entertaining possible worlds, by interpreting the hypothetical scenarios as single events. Counterfactual thinking, however, develops after the preschool years, as indicated by the difficulty of 3-5-year-olds to understand open counterfactual questions.

Rafetseder and Perner (2010a) explored whether children's correct answers to counterfactual questions reflect their ability to reason counterfactually. They hypothesized that younger children treat subjunctive questions as indicative, and they give realist responses to counterfactual questions. They used the example from Harris, German, and Mills (1996), in which children watch a doll walking in a house with dirty shoes and they are being asked a counterfactual question (57a).

(57) a. "What if Carol had taken her shoes off—would the floor be dirty?".
b. "If Carol takes off her dirty shoes and walks without shoes over the floor, is the floor dirty?".

Children gave the right answer ("clean") to the counterfactual question. However, Rafetseder and Perner (2010a) claimed that children's answer does not prove that children reason counterfactually, because the answer would be the same if children were asked the same question in indicative (57b). Children's answers showed that they succeed with imagined extensions of real events, and thus they can think of future hypothetical scenarios. The contrast of future hypothetical reasoning to counterfactual reasoning is that the first extends a single actual event in the future, whereas the second creates a parallel scenario that is reality dependent.

Counterfactuality is indicated by the use of subjunctive. The authors conducted an experiment to test their hypothesis about realist errors in counterfactual questions that would support the claim that children answer correctly in counterfactual questions, because they treat subjunctive as indicative. They presented children with a scenario of a doctor being at a hospital and Jack, a young boy, getting hurt in the swimming pool of his house. The doctor was informed that Jack needed help and he went to the swimming pool to help him. Children were asked 'where the doctor would be, if Jack hadn't hurt himself'. The results showed that children gave realist responses to the counterfactual question, by answering where the doctor actually was. To conclude, children can relate counterfactual to actual situations systematically after the age of 6, when counterfactual emotions emerge. Before the age of 6, children use basic conditional and not counterfactual reasoning (Rafetseder, Christi-Vargas and Perner 2010). When counterfactual and basic conditional reasoning questions have the same answer, then younger children respond correctly. Younger children use basic conditional reasoning to give right answers to questions that are in contrast with reality and counterfactual reasoning is not necessary to do so. The ability of children to entertain counterfactual situations, is not the same as the ability to reason counterfactually (Rafetseder, Christi-Vargas and Perner 2010; Rafetseder and Perner 2010).

4.2.3 Counterfactual reasoning and counterfactual emotions.

Counterfactual reasoning is the mental representation of how the world would be in the present, had things been different in the past. Lewis (1973) described counterfactual reasoning as the "nearest possible world", that is the ability to create an imaginative world that is as close to the actual world as possible. The counterfactual emotions that arise from counterfactual thinking are regret and relief. The development of regret is a matter of controversy, as some researchers (Weisberg & Bech 2012) suggest that it has developed at the age of 4 to 5 years old, whereas others (Rafetseder & Perner 2012) suggest that regret develops at about the age of 9.

Children first develop the ability to think and talk about what they can observe in the present moment of utterance. Counterfactual reasoning is related to the feeling of regret. Regret is a counterfactual emotion (Kahneman & Miller 1986) that emerges when a person can compare reality to its counterfactual alternatives.

Beck et al. (2006) suggested that understanding open counterfactuals is the same process as understanding regret, as it requires to think of alternative events as possibilities and to think of a point back in time where either event could have happened. In a consequent study, Beck and Crilly (2009) claimed that children can understand counterfactual conditionals before experiencing regret.

Weisberg and Beck (2010) conducted two experiments to test children's thinking about regret and relief. In the first experiment, 31 children participated in a decision-making game. They had to choose between two boxes that contained stickers. One of the two boxes contained more stickers than the other. Children rated their happiness before and after learning the content of the box they did not choose. Children showed regret at the age of 5 and relief at the age of 7. In the second experiment, some children played the same game, and some other observed other participants to play the game (53 children in total). The children who played the game gave the same responses as in the first experiment. The children who rated other participants' happiness, however, could not predict the counterfactual feelings of others, at least before the age of 7. This finding shows that children do not start experiencing and understanding regret and relief simultaneously.

The fact that children did not experience regret or relief before the age of 5 is also supported by Guttentag and Ferrell (2004) who claimed that children's emotions are not affected by counterfactual scenarios at the age of 5. Before the age of 9, children experience the feeling of regret based on what they got and not of what they could have got (Rafetseder & Perner 2011).

Rafetseder and Perner (2011) explored the reason why children start developing counterfactual reasoning at the age of 3 but start experiencing the counterfactual feeling of regret after the age of 6. To answer to this question, they conducted four experiments. The participants in experiment 1 were 37 children from 3 to 6 years old. They had to choose between two boxes, and they could keep what was inside the box of their choice. The children were asked to rate their feelings in a dependent and an independent condition, by choosing how happy they were in a scale with smiley-faces. In the dependent condition, they had to say 'how happy they are' before (baseline question) and after (test question) they saw the content of the box they did not choose. In the independent condition, the participants were asked the same question only after they had seen the content of the unselected box. In other words, they were only asked the test and not the baseline question. The lower the response to the question, the higher the regret of children. The results showed that children's satisfaction with what they got (a candy for example) was not altered when they saw the content of the box they did not choose (five candies). Only a few 6-year-old children (at best 38%) seem to experience regret.

The question of what would happen with adults led to experiment 2. 39 adults participated in the second experiment. The procedure was the same, but the content of the boxes was a certain amount of money (0,5 or 50 euros) instead of candies. Adults showed regret in contrast with children. The question of when children start to show regret in the independent task, and the question of whether the significant difference between children's answers in the dependent and the independent questions is due to pointing out what they could have got, or the double questioning led to the third experiment.

In experiment 3, 98 children from 3 to 14 years old had to choose between two identical boxes that contained items that were different either in quantity (one or many candies) or quality (attractive or unattractive stickers) and then they had to scale their happiness in a numbered bar to avoid the choice of the smiley-face children liked the most. Children from 4 to 8 years old did not express regret as shown from their answers in the baseline and the independent test question. The results also showed a difference between the baseline and the independent test question and not between the dependent and the independent test question. This means that children changed their responses because they were asked the same question twice time and not because they were given a hint of counterfactuality.

In experiment 4 they tested whether children's use of the emotional scale does not reflect their feeling of regret. 17 children, from 6 to 8 years old, had to choose between two boxes with the same contents as in experiment 3. After they chose a box, they were asked to show how happy they are on the scale. After they gave a response, they were asked to show how happy they would be if they would have taken something better. The results showed that children were happier when they got the item that was more attractive than its alternative. They were also happier with an imaginative better alternative than a less attractive real item they got. This means that their choices on the scale reflected how happy they are when they get an item that they find attractive or unattractive.

Children do not experience counterfactual regret before the age of 9. This conclusion strengthens the hypothesis of the authors that children do not reason counterfactually before that age. It has been suggested that children have difficulty with counterfactual tasks because they cannot understand the nearest possible world constraint. Younger children treat a counterfactual scenario as a new situation, and they interpret it biconditionally. Only after the age of 6, children start to develop both a full understanding of counterfactuality and experiencing negative counterfactual emotions like regret (Rafetseder and Perner 2014). However, counterfactual reasoning does not develop fully before the age of 12 years (Rafetseder, Schwitalla & Perner 2013). Even though younger children give correct responses to counterfactual questions, it is suggested that it is not due to counterfactual reasoning, but due to the interpretation of subjunctive as indicative (Rafetseder and Perner 2010) or to world knowledge (Rafetseder and Perner 2011).

Beck & Riggs (2014) wrote a review of developmental studies on real world counterfactuals, that is past events that would have been different. They focused on the development of counterfactual thinking and the importance of the relation of counterfactuals to the real world. They reviewed studies on counterfactual emotions and concluded that researchers do not agree on when children start experiencing regret. Even though some researchers claim that children experience regret at the age of 4-5 (Weisberg & Beck 2012), others suggest that they experience that feeling after the age of 6 (O'Connor, McCormack & Feeney 2014) or even later, after the age of 9 (Rafetseder & Perner 2011). Counterfactual reasoning develops over a prolonged period due to various factors, like language, domain specific knowledge, children's perception of time and executive processes like inhibition, working memory and attentional flexibility.

Rafetseder and Perner (2014) reviewed studies on the development of counterfactual reasoning and showed that it is linked to the expression of counterfactual emotions that are investigated by social (regret, relief, disappointment, remorse, blame), clinical (depression, social anxiety, procrastination, Parkinson's disease, schizophrenia) and cognitive developmental psychology (ability to infer another person's false belief that is important for social understanding) . The authors referred to Lewis's nearest possible world constraint as one of the main characteristics of counterfactual reasoning. It describes how the world would be if some features of the actual world would change but some others were left the same.

O'Connor, McCormack and Feeney (2014) examined regret and decision making in young children and they found that experiencing regret and adaptive choice switching are strongly related. The ability to experience regret and make a decision after a bad outcome of a past decision emerge simultaneously. They also suggested that experiencing regret helps children's decision making. Counterfactual thinking is ultimately beneficial for people, even though it can be both harmful and beneficial in the beginning (Roese 1997).

4.3 Summary

The development of wishes in this chapter is explored through the findings of prior studies on the development of counterfactuals, modality, counterfactual reasoning, and counterfactual emotions. In their corpus-based studies, Tulling and Cournane

(2019, 2022) found that children make tense errors as they use present instead of past and that counterfactual wishes emerge before counterfactual conditionals.

In past counterfactuals, children make the same error by using past simple instead of pluperfect (Crutchley 2004). Children's difficulty with the right use of tense in counterfactual constructions is due to their lack of understanding the modal use of past that denotes counterfactuality (Rouvoli, Tsakali & Kazanina 2019) and they are able to master counterfactual constructions only after preschool years (Beck et al 2006). Children also have difficulty to differentiate between real and unreal information (Gómez-Sánchez, Ruiz-Ballesteros & Moreno-Ríos 2020) and they start learning the meanings of the verbs with a bias towards actuality (de Villiers 2005; Kazanina, Baker & Seddon 2020). They also interpret subjunctive counterfactual questions as indicative (Rafetseder & Perner 2010), a finding that shows that they perceive what is said as real. Counterfactual thinking develops after basic conditional reasoning and children develop basic conditional reasoning at about 6 (Rafetseder, Christi-Vargas & Perner 2010). Counterfactual reasoning starts developing after 6 (Rafetseder and Perner 2014) and it is assumed that it is not fully developed before the age of 12 (Rafetseder & Perner 2011; Rafetseder, Scwitalla & Perner 2013). The development of counterfactual feelings requires the understanding of counterfactuals, and thus the development of counterfactual reasoning precedes the development of regret, which takes place at 4 to 5 and for some researchers at the age of 9 years old (Beck et al. 2006, Beck & Crilly 2009, Rafetseder & Perner 2011).

Younger children prefer what is real, and thus interpret the subjunctive as indicative and use present tense to denote non-actuality in the present, as they have not acquired the counterfactual use of 'fake' past. There is a general agreement that the development of counterfactual reasoning and counterfactual structures gets better with age (Crutchley 2004; Beck et al. 2006; Beck & Riggs 2014).

The prediction about the development of unattainable/ counterfactual desires is that younger children, i.e. at least before the age of 6, have not yet acquired the use of 'fake' past, subjunctive mood and irrealis modality. In a comprehension study of unattainable desires/ wishes children are expected to treat counterfactual desires as attainable (otherwise, *wishes* as *wants*) and thus to interpret X-marked desires as O-marked. In a production task children are expected to make tense errors, in accordance with the previous studies.

5 The study: the development of wishes in Greek

5.1 Questions

This study aims to explore at what age children acquire attainable and unattainable desires. The main questions this topic raises are the following:

1. At what age do children distinguish between O-marked and X-marked desires?
2. At what age do children understand that pluperfect denotes counterfactuality?

Younger children are expected to have an adultlike performance with O-marked desires, meaning that they understand them correctly. With X-marked desires, they are expected to fail or at least give less correct responses. It is also predicted that older children will give more correct responses to X-marked desires than younger children. The older children get, the less errors they are expected to make. We thus hypothesize that children first acquire O-marked desires, and they younger they are, the more errors they make by misinterpreting X-marked desires as O-marked. This misinterpretation is predicted to be realized through present for past tense errors.

5.2 Methodology

5.2.1 Participants

The participants of the study were 30, 4–9-year-old (mean age: 7;7), all typically developing monolingual Greek speaking children living in Crete. They were divided into Group 1 (mean age: 6;4) and Group 2 (mean age: 8;9). The split was made due to the point we noticed there is a developmental shift in children's responses All participants are typical monolingual Greek speaking children, living in Crete. The parents of the children were approached through personal communication, from a greater circle of relatives and friends. The answers of a child with learning difficulties were excluded from the analysis of the results.

The control group consisted of 30 adults, 20 to 56 years old (mean age: 34).

All participants are native speakers of Greek living in Greece, and they were approached in the same way as the parents in the main group.

5.2.2 Experimental design and procedure

A comprehension picture-matching task, in a 'Tell-me-who-is-talking' mode (PCIbeX platform), was developed to test children's comprehension of O-marked and X-marked desires[11]. The experiment was built with Penn Controller for Internet Based Experiments (PCIbeX) (Zehr & Schwarz 2018). PCIbex is the extension of IbeX, a platform for behavioral experiments based on JavaScript programming language. PCIbex Farm is the web platform, where experiments are designed through guided programming language, with preview possibility, and are shared through different URLs as Open Science resources, and with the participants for data collection (Schwarz & Zehr 2021). The

[11] I would like to thank Brian Dillon for our discussion on the experimental design and for introducing me to PCIbeX and Roman Feiman for motivating me to look at the experimental items from a child's perspective when building up an experiment.

materials used in the design are illustrations of colored creatures[12] and animals[13] and voice recordings[14].

The idea is that children play a game that is based on a 'Festival of Desires', that takes places in a magical world. According to the experimental procedure, the child was told a short story about little creatures who went to the Festival of Desires. During the Festival, the little creatures can become any colour they want. Then children are informed that they will listen to the 'interviews' the little creatures gave, talking about the colour they got and if they want to remain the same or if they would like to be different. Children would watch on the screen two little creatures, while listening to an utterance and they have to tell the experimenter which creature is talking and click on the icon.

Children were told that all they had to do to 'play the game', was to listen to the sentence and choose the creature or animal that is talking. The 'Tell me who is talking' task,[15] was presented as an interesting and easy game. No matter what answers they gave, children were told that they were doing great. They were also told that if they participate, they would help the experimenter, resulting in strengthening of children's willingness for participation.

The experimental procedure started with the experimenter informing the parent/ guardian of the child (for the experimental group) or the adult for the (control group) about the study and giving them the consent form to read it, fill in the information needed and sign it. The consent form was given to the participants either printed or through a link in the online experiment.

The link to the experiment for children is different from the link to the experiment for the control group of adults. The two experiments are identical. A difference between the experiments, was that the experimental child group had to sign a different consent form than the control group of adults

During the experimental process the experimenter used a laptop or a phone. For child participants, the experimenter presented the study as an online game that is based on a story about the 'Festival of Desires'. Adults were told that they will take part in an online experiment.

The first thing the participants saw when they opened the link was information about giving consent, the link to the consent form, and the email where they would send it signed. Participants were asked to click on a button if they agreed to participate so that the experiment would start.

The second thing shown on screen was the story the experiment is based on. Adults read the story to understand what they had to do. Children were told the story by the experimenter (or their parent after being instructed by the experimenter) that is based on the following text[16] that appears on screen.

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[12] The little creature illustrations used in this experiment were created by Izoldie Tsagkaraki.

[13] The PNG images of animals were retrieved from: https://freepngimg.com/, where all images are distributed with license (https://creativecommons.org/licenses/by-nc/4.0/) No changes were made.

[14] The voice recordings used in this experiment were provided by the author.

[15] The name of the task (*'Tell-me-who-is-talking'*) was proposed by Vina Tsakali in one of our discussions on the experimental design.

[16] The original text in Modern Greek: "Καλωσήρθες στη Γιορτή των Επιθυμιών! Εδώ θα συναντήσεις μερικά ζωάκια και χρωματιστά πλασματάκια που πηγαίνουν στην Γιορτή των Επιθυμιών. Σε αυτή τη γιορτή, τα πλασματάκια και τα ζωάκια μιλάνε άλλες φορές για κάτι που θέλουν να μείνει ίδιο και άλλες φορές για κάτι που θα ήθελαν να είναι διαφορετικό.
Διαδικασία: Κάθε φορά που θα ακούς ένα πλασματάκι ή ζωάκι να μιλάει, θέλω να βρεις ποιο από τα δύο είναι αυτό που μίλησε κάνοντας κλικ πάνω του!".

"Welcome to the Festival of Desires! Here you will meet some animals and colored creatures that go to the Festival of Desires. In this celebration, little creatures and animals talk sometimes about something they want to remain the same and other times about something they would like to be different.

Procedure: Every time you hear a creature or animal talking, I want you to find out which of the two is the one that spoke by clicking on it!"

After the (code) name of the participant was filled in the box at the bottom, the 'game' began. The first three sentences were familiarization trials, and they were not included in the results analysis. After the participants answered to the three questions, they clicked on a button where the following sentence it is written: "Now that you have learnt how to play this game, let's start!".

Every time children listened to a sentence, they saw either two creatures or two animals and they had to find which of the two uttered the sentence. Children answered either by telling who said the sentence and clicking on it or by telling who said the sentence so that the experimenter clicks on the corresponding picture. It depended on what the participant prefered each time. The experimental items appeared in random order. The images also appeared in random order, either at the left or the right side of the screen. The experiment stoped automatically, when the participant answered to all the items and a 'thank you' message appeared on screen. The experimenter thanked children for participating, after 'playing the game' and gave them out stickers as a reward.

5.2.3 Conditions and items

In total, there are 8 conditions. There are 4 critical conditions with 6 items each and 4 control conditions with 6 items each. There are also 3 familiarization items. In total, there are 51 items in the experiment. Critical conditions aim to test how children understand O-marked in comparison to past X-marked desires, as well as if the lexical meaning of the desire-verb affects their choice. Conditions 1 and 2 tested O-marked desires in combination with the verbs *thelo* ('I want') and *efxome* ('I wish'). Control conditions aim to distract children from the desire-sentences in order to avoid biased responses. Control conditions are *glad/ like*-sentences formed with *mu aresi* and *tha mu arese*.

5.2.3.1 Critical items

Critical items are used to test O-marked and X-marked desires. Conditions 1 and 2 test O-marked desires and Conditions 3 and 4 test X-marked desires, as shown in Table 4.

Conditions of critical items	Linguistic stimuli	Candidates (in random order)	Who is talking	Figure
Condition 1: Want/O-marked	Thelo na mino kocino! 'I want to remain red.'	blue + red	Red	4
Condition 2: Wish/O-marked	Efxome na mino kocino! 'I wish to remain red.'	green + red	Red	5
Condition 3: Want/past X-marked	Tha ithela na ixa mini mov! 'I wish I had remained purple.'	orange + purple	Orange	6

Condition 4: Wish/past X-marked	Efxome na ixa mini mov! 'I wish I had remained purple.'	blue + purple	Blue	7

Table 4 *Critical Items Conditions*

Condition 1 tests *thelo* ('I want') *O-marked desires*. Condition 2 tests *efxome* ('I wish') *O-marked desires*. The complement in both conditions is subjunctive and the verb in the complement is future perfective. The two conditions differ in the lexical meaning of the verb of the matrix clause. The verb *thelo* is used in Condition 1, whereas the verb *efxome* is used in Condition 2. O-marked desires appear as complements of both these verbs to check whether children interpret the two sentences in a different way. The difference in the lexical meaning of the desire verbs should not affect the answers of the participants, as being discussed in the predictions section. In both conditions, a colored creature expresses the desire to remain the same. A red creature for example, can express the desire to remain red, like in Figures 4 and 5.

In Condition 1, as shown in Figure 4, the O-marked desire sentence *'Thelo na mino kocino'* ('I want to remain red') is being heard while a blue and a red creature are shown on screen. The correct response in the *'Tell-me-who-is-talking'* task is the 'red creature'. Similarly, in Condition 2, as shown in Figure 5, the O-marked desire sentence *'Efxome na mino kocino'* ('I wish to remain red') is being heard while a green and a red creature are shown on screen. The correct response in the task is, again, 'the red creature'.

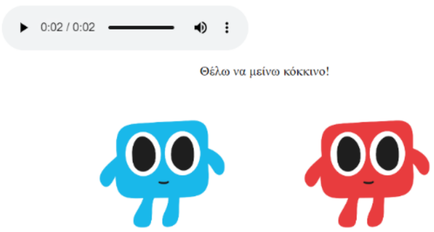

Figure 5 Condition 1: Want/ O-marking

Figure 4 Condition 2: Wish/ O-marking

Conditions 3 and 4 are X-marked desire sentences. Condition 3 tests *tha ithela* ('would want') X-marked desires and Condition 4 tests *efxome* ('wish') X-marked desires. In both conditions, the complement is subjunctive, and its verb is in pluperfect. The two conditions differ in the lexical meaning of the verb of the matrix clause, but again children are expected to interpret them in the same way and either succeed or fail in both conditions.

Figure 7 Condition 3: Would want/ X-marking

Figure 6 Condition 4: Wish/ X-marking

Figures 6 and 7, show how Conditions 3 and 4 are experimentally tested. The Figures show that a purple creature cannot wish it had remained purple, because it comes in contrast with its purpleness, i.e. the color that it already has.

In Condition 3, as depicted in Figure 3, the X-marked desire '*Tha ithela na ixa mini mov'* ('I wish I had remained purple') is being heard while an orange and a purple creature are shown on screen. The correct response to the *'Tell-me-who-is-talking'* task is 'the orange creature'. In Condition 4, as depicted in Figure 4, the X-marked desire *'Efxome na ixa mini mov'* ('I wish I had remained purple') is being heard while a blue and a purple creature are shown on screen. The correct response to the *'Tell-me-who-is-talking'* question is the blue creature.

The stative verb *meno* ('I remain') is used is all 24 critical items. The initial idea was to use the dynamic verb *ɉinome* ('I become') instead. Children develop a dynamic verb meaning earlier than a stative verb meaning (Herr-Israel & McCune 2006). For this reason, a dynamic verb seems to be a better choice than a stative verb. However, with the dynamic verb *ɉinome,* it would not be possible to differentiate between O-marked and X-marked desires, because the creature or animal chosen would have to be the same[17]. With *meno,* this problem is solved, as depicted in Table 5.

Meno* vs *ɉinome	**Linguistic stimuli**	**Candidates**	**Who is talking**	**Does the answer reflect the difference between O-marking and X-marking?**
O-marked *become*	Thelo na ɉino prasino! 'I want to become green.'	green +red	Red	No
X-marked *become*	Tha ithela na ixa ɉini prasino! 'I wish I had become green.'	green + red	Red	No
O-marked *remain*	Thelo na mino prasino! 'I want to remain green.'	green + red	Green	Yes
X-marked *remain*	Tha ithela na ixa mini prasino! 'I wish I had remained green.'	green + red	Red	Yes

Table 5 *The reason why remain is selected over become*

5.2.3.2 Control items

In this study we use control items with the role of fillers[18]. Fillers aim at distracting the participants from focusing on the form of critical items. Control items aim at confirming that the answers to the experimental questions are not random but reflect the participants' level of understanding. To avoid biasing participants, it has been recommended to follow a neutral ratio of critical and filler sentences in the experimental design, like 1:1. For children, the ratio can be smaller, as they are less likely to be thinking on the study's aim (Schmitt & Miller 2010, Marinis & Cunnings 2018, Arehalli & Wittenberg 2021). Following the critical-filler item ratio, the number of control items is equal to the number of critical items. Control items have a similar structure to critical items as well. The 24 control items of the experiment are divided in 4[19] conditions as shown in Table 6.

[17] I would like to thank Marina Mastrokosta for pointing out that *ɉinome* would give the same answer for both O-marked and X-marked desires.
[18] I would like to thank Despina Oikonomou for our discussions on the conditions of the experiment and especially for her comments on the control items.
[19] For avoidance of confusion: Conditions 1-4: critical item conditions, Conditions 5-8: control item conditions.

Conditions of control items	Linguistic stimuli	Candidates (in random order)	Who is talking	Figure
Condition 5: I like/ O-marked	Mu aresi pu ime mov! 'I like being purple!'	snake + hippo	Hippo	8
Condition 6: I would like/ present X-marked	Tha mu arese na imun roz! 'I would like to be/ have been pink!'	giraffe + flamingo	Giraffe	9
Condition 7: I like/ O-marked	Mu aresi pu niaurizo! 'I like that I meow!'	cat + giraffe	Cat	10
Condition 8: I would like/ present X-marked or FLV	Tha mu arese na petusa! 'I wish I was flying!'	elephant + colibri	Elephant	11

Table 6 *Control Items Conditions*

Condition 5 uses *mu aresi pu ('I like that') O-marked sentences*. The O-marked complement is present imperfective. Condition 6 uses *tha mu arese na* ('I would like to') X-marked sentences. It should be noted that the complement of *Tha mu arese O-marked sentence* is not ambiguous, even though it is in past imperfective. It is a present X-marked sentence.

The reason why Conditions 5 and 6 are presented together, even though the first is about an O-marked and the second about an X-marked sentence, is because the embedded verb in both cases is stative and it expresses color-identity. Conditions 5 and 6 are depicted in Figures 8 and 9 respectively.

In Condition 5, two animals appear and one of them talks about liking the color it has. In Figure 8, the O-marked utterance *mu aresi pu ime mov* ('I like being purple') is being heard while a green snake and a purple hippo are shown on screen. The correct response to the *'Tell-me-who-is-talking'* task is 'the hippo'.

In Condition 6, two animals appear and one of them is saying that it would like to have a different color. In Figure 9, the present X-marked utterance *'tha mu arese na imun roz'* ('I would like to be pink') is being heard while a yellow-brown giraffe and a pink flamingo are shown on screen. The correct response to the task is 'the giraffe'.

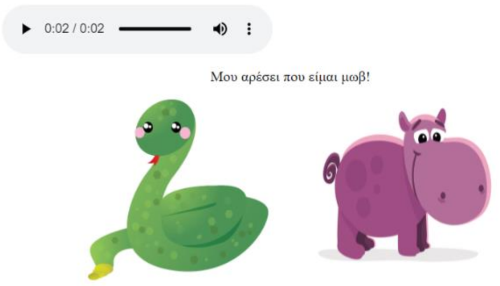

Figure 9 Condition 5: Like/ O-marked predicate

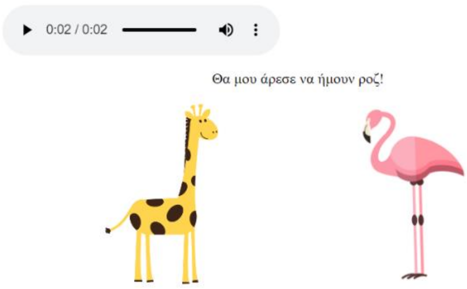

Figure 8 Condition 6: Would like/ X-marked predicate

Conditions 7 and 8 differ from Conditions 5 and 6 in that the embedded verb expresses activity and not identity. Condition 7 uses *mu aresi pu ('I like that') O-marked sentences*, with a dynamic verb in present imperfective. Condition 8 uses with *tha mu arese ('I would like') FLV or X-marked sentences*, with a dynamic verb in past imperfective. The difference between conditions 6 and 8 is that the past imperfective form of a dynamic verb yields an ambiguity between an X-marked and a FLV scenario.

In Condition 7, two animals appear on screen and one of them talks about liking a certain ability that it has. In Figure 10, the O-marked utterance *'mu aresi pu niaurizo'* ('I like to meow') is being heard while a cat and a giraffe are shown on screen. The correct response to the *'Tell-me-who-is-talking'* task is 'the cat'.

In Condition 8, two animals appear on screen and one of them is saying that it would like to have an ability that it does not have. In Figure 11, the FLV/ present X-marked utterance *'tha mu arese na petusa'* ('I would like to fly/ have been flying') is being heard, while an elephant and a colibri are shown on screen. Both answers are correct in this condition. If the interpretation is X-marked the participant should choose the elephant, whereas if the interpretation is O-marked, the participant should choose the colibri.

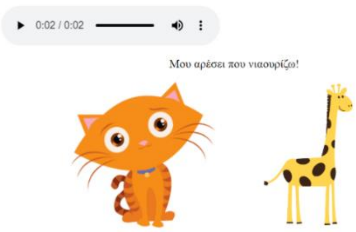

Figure 11 Condition 7: Like/ O-marked activity

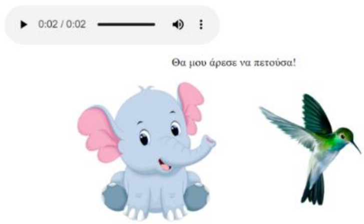

Figure 10 Condition 8: Would like/ X-marked activity

Control items are similar to critical items, but not identical to them. The reason why the verb *mu aresi* ('I like') was chosen for the critical item sentences, is that it is a propositional attitude like the verbs *thelo* and *efxome*. *Mu aresi* expresses a positive attitude towards a proposition. Both mu *aresi* and *thelo/ efxome* contain the attitude of preference and they can take O and X-marked CPs as complements. If *I-like* sentences have the same semantics as *be glad* sentences, then they are used to express that the attitude holder desires some worlds that considers as true, more than others. Wishes are about desires in worlds that the attitude holder does not consider as true and wants are about desires that are possible, but not guaranteed, like in be glad sentences (Grano 2021).

Both control and critical items have embedded stative complements, whereas only control items have embedded dynamic complements in Conditions 7 and 8. Critical X-marked items are in pluperfect, whereas control X-marked items are in past imperfective. In Conditions 5 and 6, the embedded verb is stative and in this specific context it is necessarily interpreted as counterfactual.

In Conditions 7 and 8, the embedded verbs *walk, fly, swim, run, meow* and *bark* can yield ambiguity that depends on the context. In Condition 7, there is no ambiguity because the indicative marker *pu* makes it possible only for the cat to say that it likes the fact that it meows. In Condition 8 there is ambiguity. If an elephant says that it would like to fly, it means that it would like to have the ability to fly. The verb in this scenario is generic. If a colibri says that it would like to fly, it means that it has the ability to fly and it would like to start doing it. The verb in this scenario is dynamic.

If the verb is interpreted as generic, then it is more possible that past imperfective yields a present X-marking meaning. If the verb is interpreted as dynamic, then it is more possible that past imperfective yields a FLV meaning. The fact that it is the control and not the critical items that have ambiguity frees us from the concern of confusing the participants. Their answers are predicted to follow a specific pattern and if they do so, they will choose readings on ambiguous sentences in a predictable way too.

5.3 Predictions

Answering to the questions is Table 7 leads to the predictions that follow the research hypothesis, as well as the findings of previous research and theory.

	Questions behind predictions
Question 1	Do all participants understand (conditions 1,2, 5 and 7) O-marked desires?
Question 2	Does the difference in the lexical meaning of *thelo* (condition 1) and *efxome* (condition 2) affect sentence comprehension?
Question 3	Are X-marked desires (conditions 3 and 4) interpreted as O-marked or X-marked by younger children and older children?
Question 4	Are O-marked desires (conditions 1 and 2) acquired before X-marked desires (conditions 3 and 4)?
Question 5	Does the difference in tense of [+ past] *tha ithela* (condition 3) and [- past] *efxome* (condition 4) affect the answers of the participants?
Question 6	Are structures in pluperfect (conditions 3 and 4) treated as structures in past imperfective (6 and 8)?
Question 7	Past imperfectives: Are X-marked predicates (condition 6) and X-marked activities (condition 8) interpreted in a different way? What is the counterfactuality/ X-marking percentage between the two?

Table 7 *Questions behind predictions*

The first prediction is that O-marked desires are interpreted as O-marked by all participants. This means that all children are expected to answer correctly to the items in Conditions 1, 2, 5 and 7.

The second prediction is that *thelo* and *efxome* O-marked sentences are interpreted in the same way by all participants. Again, all children are expected to answer correctly in Conditions 1, 2, 5 and 7. This prediction follows from the first, but there is reason to present them separately. The reason is that if the lexical meaning of the verbs *thelo* and *efxome* gives rise to different interpretations of the condition, then children are expected to also answer in a different way.

The third prediction is that X-marked desires are interpreted as O-marked by all younger children (Beck et al. 2006, Rafetseder; Christi-Vargas & Perner 2010; Rouvoli et al. 2019). Children that are older than 6, have started developing counterfactual understanding and they are predicted to answer correctly at least in some of the X-marked items. After the age of 8-9, when children have developed counterfactual thinking and the counterfactual emotion of regret (Guttentag & Ferrel 2004; Beck & Crilly 2009; Rafetseder & Perner 2011; Weisberg & Beck 2010, 2012), they are expected to interpret almost all X-marked desires as X-marked.

The fourth prediction, that follows from predictions 1 and 3, is that O-marked desires are acquired before X-marked desires. This prediction is following the findings of previous studies according to which realis structures about the actual world are acquired before irrealis counterfactual structures (de Villiers 2005; Beck et al. 2006; Weisberg & Beck 2010; Kazanina et al. 2020; Gómez-Sánchez et al. 2020).

The fifth prediction is that the difference in tense of the past tense form *tha ithela* (Condition 3) and the present tense form *efxome* (Condition 4) does not affect the answers of the participants, in the same way that the lexical meaning of the two verbs is not expected to make a difference. Both conditions are expected to be treated in the same way by both children and adults.

The sixth prediction is that past X-marked structures (Conditions 3 and 4) will be interpreted in the same way with present X-marked structures (Conditions 6 and 8) by younger children who interpret them as O-marked and by children that interpret them as X-marked. The children who are in the process of acquiring X-marking, and counterfactual reasoning in general, are expected to treat only past X-marked structures as X-marked and interpret past imperfectives as O-marked.

The seventh prediction is about an internal distinction between control conditions. Past imperfective structures are found with stative and dynamic verbs (Conditions 6 and 8). In Condition 6, the embedded verb is stative and in Condition 8, the embedded verb can be either interpreted as generic/ stative or as dynamic. The X-marking interpretation is more possible with past imperfective predicates (generic/ stative verbs) than with past imperfective activities (dynamic verbs). Children are expected to give more correct answers to Condition 10 than to Condition 12.

These predictions stem from the questions that were raised due to theory. The aim is to explore the development of X-marked structures in child language and set the ground for future research.

5.4 Results

The overall results of the study are shown in Figure 12. Figures 13 and 14 present the correct responses of Group A and Group B, respectively. Figure 15 shows the overall correct responses to control items.

Figure 12 depicts the overall correct responses of both children and adults in Conditions 1-4.

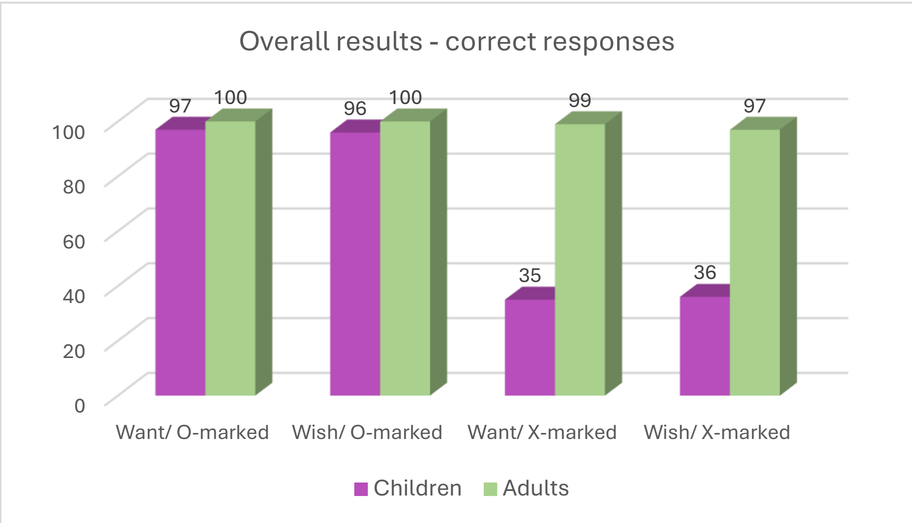

Figure 12 Overall results - critical items

In Conditions 1 and 2, the group of adults has a ceiling performance (100% correct responses). In Conditions 3 and 4 the group of adults responds correctly up to a rate of 97-99%. The fact that adults respond correct in every Condition in an almost 100% percent shows that the experiment works. The 1-3% wrong responses of adults to

Conditions 3 and 4 is an indicator that children will give less correct responses to Conditions 3 and 4 than to 1 and 2.

The group of children performs adultlike in Conditions 1 and 2, with a percentage of correct responses up to 96-97%. Children understand O-marked desires correctly, in an adultlike manner. In Conditions 3 and 4, the correct responses of children show that they are far from adultlike, with 35-36% correct responses. In X-marked desires children fail up to a rate of 64 – 65%. Children in Conditions 3 and 4 if compared to their responses to Conditions 1 and 2, but also to the responses of adults in all four Conditions. The difference between children's correct responses to X-marked and to O-marked desires is highly statistically significant, as the p-value is almost 0.

The responses of both the children and the adult group to Conditions 1 and 2 shows that they treat them equally. If the lexical meaning of the desire verbs *thelo* and *efxome* would affect the interpretation of the O-marked desires, it would affect their responses to this task. However, the responses to Conditions 1 and two are equally correct.

Similarly to O-marked desires, the responses of both children and adult groups to Conditions 3 and 4 show that both tha ithela and efxome X-marked desires are understood in the same way. All adults understand X-marked desires as X-marked, whereas most children understand X-marked desires as O-marked, irrespectively of the lexical meaning of the desire verb.

Figures 13 and 14 show the correct responses in Conditions 1 – 4 by the group of younger (Group A: 4-7;11 years old, n=15) and older children (Group B: 8-10 years old, n=15), respectively. Both groups perform adultlike in O-marked desires. In X-marked desires, none of the groups performs like adults.

Figure 13 shows that younger children give 97 – 99% correct responses to Conditions 1 and 2, whereas they fail with Conditions 3 and 4, as the correct responses can hardly reach a percentage of 20%, ranging from 17 – 19 %. Figure 14 shows that correct responses of older children to O-marked desires are identical to these of adults reaching a 100% percentage, whereas correct responses to X-marked desires range from 52 to 53%. This shows that younger children fail at a rate of 81-83%. The developmental pattern advances just before the age of 8, as the failure percentage falls at 47-48%. All in all, there is a significant difference in the number of correct responses in Conditions 3 (want/ X-marked) and 4 (wish/ X-marked) between Group A and Group B.

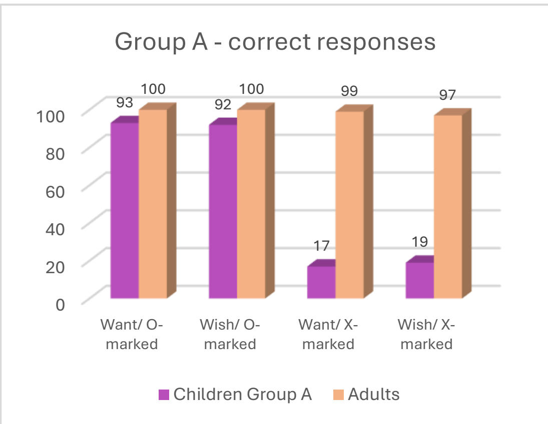

Figure 14 Group A – correct responses

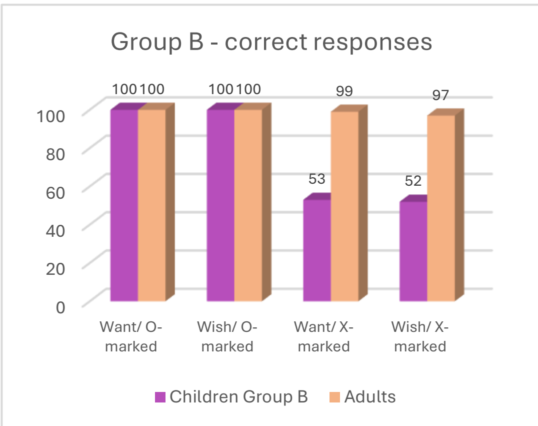

Figure 13 Group B - correct responses

Figure 15 presents the overall correct responses of all children and adults in control items. accordance with children's responses to critical items, children's correct responses to control items in Conditions 5 (like/ O-marked predicate) and 7 (like/ O-marked activity) show that *I like*-sentences with O-marked predicates and O-marked activity verbs have fully developed in child language. Children's responses to these sentences are 97-99% correct. Adults' responses in Conditions 5 and 7 are 100% correct.

On the other side, children's responses to Condition 6 (*would like/* X-marked predicate) and Condition 8 (would like/ X-marked activity/ FLV) are 43-47% correct. Adults give 99% correct responses to Conditions 6 and 8. The difference between the percentage of correct responses in Conditions 6 and 8, that is a difference between *would like* sentences with predicates (47% correct) and sentences with dynamic verbs (43% correct) might indicate that the choice of the category of the embedded verb affects children's interpretation of the sentence. This topic might arise in future work, as in this study the *like/ would like*-sentences are control items. The fact that the total responses to critical and to control items follow a similar developmental path shows that the responses to critical items were probably affected but not driven by color association. The overall results with control items do not deviate from the bigger picture as created by the results of critical items (Figure12).

Figure 15 Overall results – control items

The overall results of the study as depicted in Figure 15 confirm the predictions that O-marked desires are interpreted as O-marked by all groups of participants and that X-marked desires are interpreted as O-marked by the group of younger children. O-marked desires are thus developed earlier that X-marked desires. These findings are supported by the overall responses to control items as shown in Figure 15.

The graph in Figure 15 shows that children give 43-47% correct responses to present X-marked sentences, which means that they perform better with present than with past X-marked desires, where they respond correctly 35-36%. This indicates that children first acquire present X-marked desires and then past X-marked desires.

To conclude, the results confirm the predictions of this study. Children's comprehension of attainable desires is adultlike. Younger cannot distinguish attainable from unattainable desires, as they interpret X-marked desires are O-marked. Older children perform better than younger in X-marked sentences. This is due to Actuality bias (de Villiers 2005; Kazanina et al 2020), the prolonged development of counterfactual reasoning and probably because of the misanalysis of the pluperfect associated with X-marking (Crutchley 2004; Rouvoli et al 2019; Tulling & Cournane 2019, 2022).

6 Discussion

The aim of this thesis was to explore the development of (un)attainable desires in child Greek though a comprehension study. The core question is whether children distinguish between attainable and unattainable desires. The results confirm the prediction that all typically developing children interpret O-marked desires as O-marked, but not all of them understand X-marked desires as X-marked. Younger children (4-7;11 years old, n=15) fail with X-marked structures (80+% incorrect responses). Their responses show that they do not understand X-marked desires as X-marked. If children understand unattainable desires as attainable, it is because the development of counterfactual reasoning is a prolonged process and because they have a bias towards actuality.

One issue that should be discussed is whether the children that succeed with O-marked desires, but fail with X-marked desires, do so due to colour association. It might be the case that every time children listen to a colour-word, they pick the creature of the same colour. This way, they will always answer correctly to O-marked structures and incorrectly to X-marked structures. However, if children have acquired that pluperfect denotes counterfactuality, they will negate the uttered-colour creature and they will respond correctly. The children that have not acquired the counterfactual meaning of the X-marked desire, will interpret the event as something that happened and not as something that did not happen and they will choose the creature that matches the colour uttered in the sentence.

The control items of the experiment were used to check whether children give biased answers due to colour association or not. If children answer to both critical and control items in a similar way, then their choice is not affected by the colour uttered in the sentence, but by their comprehension of the structure. The results showed that children treat both critical and control items depending on whether the sentences are O or X-marked. The children that treated X-marked desires as O-marked, also treated the X-marked sentences in control items as O-marked.

The control item sentences are about animals who like the abilities that they have or would like to have, i.e. a cat versus a dog being able to meow. Even though they were not part of the main experiment, control items were not selected aiming only to distract children from getting biased. They tested children's responses to *(would) like*-sentences, as well as their understanding of past imperfective structures setting the base for the study of the development of present X-marked desires, as well as of *be-glad* sentences.

The assumptions about the order of the development of desire structures as presented in Table 9 are shaped by children's responses to control items, where they seem to perform better with past imperfectives than with past X-marked desires in pluperfect.

Order of development	Types of desires	Morphology of complement
1	O-marked desires	not past + subjunctive
2	Present X-marked desires + identity	past imperfective + subjunctive
3	Present X-marked desires + activity	past imperfective + subjunctive
4	Past X-marked desires	pluperfect + subjunctive

Table 8 Assumptions on the order of development of desires

O-marked desires are acquired first, because they do not require deviation from the world of the speaker. Past imperfective desires can either be ambiguous and yield an O-marked (FLV) or an X-marked meaning or have a stricter counterfactual meaning with identity/ stative verbs. At the point where X-marking with past imperfective is available to children, it is expected that children will first interpret the strict X-marked as X-marked and then the ambiguous desire as X-marked. The indicator for this prediction is the percentage of correct responses to *would like* with past imperfective stative verbs (47%) that is higher than that of correct responses to *would like* with ambiguous past imperfective activity/ dynamic verbs (43%). Past X-marked desires are expected to emerge last. When children develop the meaning of pluperfect as counterfactual, they have acquired past X-marked desires.

This study is a first step towards the exploration of the development of (un)attainable desires. Recruiting more participants in the study will help to have more homogeneous age groups and to make comparisons that are not obvious with the two-group division. Analyzing the results with statistics will also help to make a deeper and more accurate evaluation of the data that will lead to the most effective presentation of the results.

Rerarding the experimental design, it is important to show that when children respond correctly to O-marked desire items, they do so because they understand that the desire is O-marked. This can be achieved through adding the condition '*I want to become/ O-marking*'.[20] Even though the verb *ɟinome* ('I become') does not work in the experimental design for X-marked desires, it can show when children respond correctly to O-marked desires. Both sentences *'thelo na ɟino kocino'* ('I want to become red') and *'thelo na mino kocino'* ('I want to remain red') are O-marked and children's responses will show whether they make a choice based on color or on the meaning of the sentence. Children's understanding of O-marking will be tested as they should not go for the same strategy when the click on the icon.

Summarizing the findings of the study, O-marked desires are acquired before past X-marked desires. *Wish* and *want*-sentences are treated in the same way, as long as their complement is the same. Attainable desires are understood as attainable by all children, whereas unattainable desires are understood as unattainable by younger children. The older children get, the more correct responses they give to X-marked structures.

The main conclusion of this study is that younger children interpret past X-marked desires as O-marked. This finding is compatible with the Actuality-bias hypothesis in

[20] I would like to thank Bergül Soykan for making this point.

language learning, suggesting that children construct verb meanings based on actual events before including modal components (de Villiers 2005; Kazanina et al. 2020). The development of X-marked desires is undoubtedly a prolonged process. It might be even more prolonged than the development of counterfactual conditionals.

Hopefully, this study will continue with the exploration of the development of X-marked desires in comparison to X-marked conditionals. The development of present X-marked desires in comparison to past X-marked desires, as well as the development of *be-glad* versus *want/wish*-sentences are further questions to be explored in future work. Further than that, the development of expressive in comparison to descriptive wishes, that is related to the EX-operator that exists in expressive and exclamatory utterances, might give rise to interesting questions regarding the exploration of speech acts. For now, these remain ideas for future work.

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Appendices

Appendix A: Critical items

Conditions	Items		Linguistic stimuli	Candidates (in random order)	Who is talking
Condition 1: Want/O-marked	**1.1**		Θέλω να μείνω πράσινο. 'I want to remain green.'	green +grey	Green
	1.2		Θέλω να μείνω ροζ. 'I want to remain pink.'	pink + yellow	Pink
	1.3		Θέλω να μείνω πορτοκαλί. 'I want to remain orange.'	orange + pink	Orange
	1.4		Θέλω να μείνω μωβ. 'I want to remain purple,'	purple + red	Purple
	1.5		Θέλω να μείνω κίτρινο. 'I want to remain yellow.'	yellow + brown	Yellow
	1.6		Θέλω να μείνω κόκκινο. 'I want to remain red.'	red + blue	Red
Condition 2: Wish/O-marked	**2.1**	1.7	Εύχομαι να μείνω καφέ. 'I wish to remain brown.'	brown + green	brown
	2.2	1.8	Εύχομαι να μείνω μπλε. 'I wish to remain blue.'	blue + red	blue
	2.3	1.9	Εύχομαι να μείνω μωβ. 'I wish to remain purple.'	purple + yellow	purple
	2.4	1.10	Εύχομαι να μείνω ροζ. 'I wish to remain pink.'	pink + yellow	pink
	2.5	1.11	Εύχομαι να μείνω πράσινο. 'I wish to remain green.'	green + purple	green
	2.6	1.12	Εύχομαι να μείνω κίτρινο. 'I wish to remain yellow.'	yellow + blue	yellow
Condition 3: Want/X-marked (PLUPERFECT)	**3.1**	1.13	Θα ήθελα να είχα μείνει κόκκινο. 'I wish I had remained red.'	red + green	green
	3.2	1.14	Θα ήθελα να είχα μείνει πορτοκαλί. 'I wish I had remained orange.'	orange + purple	purple
	3.3	1.15	Θα ήθελα να είχα μείνει μπλε. 'I wish I had remained blue.'	blue + yellow	Yellow
	3.4	1.16	Θα ήθελα να είχα μείνει καφέ. 'I wish I had remained brown.'	brown + pink	pink
	3.5	1.17	Θα ήθελα να είχα μείνει πράσινο. 'I wish I had remained green.'	green + red	Red
	3.6	1.18	Θα ήθελα να είχα μείνει μωβ. 'I wish I had remained purple.'	purple + orange	orange
Condition 4: Wish/X-marked (PLUPERFECT)	**4.1**	1.19	Εύχομαι να είχα μείνει κίτρινο. 'I wish I had remained yellow.'	yellow + brown	Brown
	4.2	1.20	Εύχομαι να είχα μείνει κόκκινο. 'I wish I had remained red.'	red + pink	Pink
	4.3	1.21	Εύχομαι να είχα μείνει καφέ. 'I wish I had remained brown.'	Brown + Green	green
	4.4	1.22	Εύχομαι να είχα μείνει πορτοκαλί. 'I wish I had remained orange.'	orange + blue	blue
	4.5	1.23	Εύχομαι να είχα μείνει ροζ. 'I wish I had remained pink.'	pink + blue	blue
	4.6	1.24	Εύχομαι να είχα μείνει μπλε. 'I wish I had remained blue.'	blue + purple	Purple

Appendix B: Control items

Conditions	Control items		Linguistic stimuli	Candidates	Who is talking
Condition 5: I like/ O-marked	5.1	1.25	Μου αρέσει που νιαουρίζω! 'I like that I meow!'	cat + giraffe	cat
	5.2	1.26	Μου αρέσει που περπατάω! 'I like that I walk!'	tiger + snake	tiger
	5.3	1.27	Μου αρέσει που πετάω! 'I like that I fly!'	parrot + pig	parrot
	5.4	1.28	Μου αρέσει που τρέχω! 'I like that I run!'	horse + turtle	horse
	5.5	1.29	Μου αρέσει που κολυμπάω! 'I like that I swim!'	dolphin +dog	dolphin
	6.6	1.30	Μου αρέσει που γαυγίζω! 'I like that I bark!'	dog + crab	dog
Condition 6: I would like/ X-marked or FLV	6.1	1.31	Θα μου άρεσε να νιαούριζα! 'I would have liked/ love to meow!'	cat + penguin	penguin/ cat
	6.2	1.32	Θα μου άρεσε να περπατούσα! 'I would have liked/ love to walk!'	hippo + snake	snake/ hippo
	6.3	1.33	Θα μου άρεσε να πετούσα! 'I would have liked/ love to fly!'	colibri + elefant	elefant/ colibri
	6.4	1.34	Θα μου άρεσε να έτρεχα! 'I would have liked/ love to run!'	horse + turtle	turtle/ horse
	6.5	1.35	Θα μου άρεσε να κολυμπούσα! 'I would have liked/ love to swim!'	whale + parrot	parrot/ whale
	6.6	1.36	Θα μου άρεσε να γαύγιζα! 'I would have liked/ love to bark!'	dog + swan	swan/ dog
I like/ O-marked	7.1	1.37	Μου αρέσει που είμαι μπλε! 'I like that I am blue!'	blue + pink	blue
	7.2	1.38	Μου αρέσει που είμαι μωβ! 'I like that I am purple!'	hippo + snake	hippo
	7.3	1.39	Μου αρέσει που είμαι πορτοκαλί! 'I like that I am orange!'	fox + dolphin	fox
	7.4	1.40	Μου αρέσει που είμαι πράσινο! 'I like that I am green!'	crocodile + kangaro	crocodile
	7.5	1.41	Μου αρέσει που είμαι ροζ! 'I like that I am pink!'	pig + tiger	pig
	7.6	1.42	Μου αρέσει που είμαι κόκκινο! 'I like that I am red!'	crab + colibri	crab
Condition 6: I would like/ X-marked or FLV	8.1	1.43	Θα μου άρεσε να ήμουν μπλε! 'I would have liked/ love to be blue!'	blue + pink	pink
	8.2	1.44	Θα μου άρεσε να ήμουν μωβ! 'I would have liked/ love to be purple!'	hippo + snake	snake
	8.3	1.45	Θα μου άρεσε να ήμουν πορτοκαλί! 'I would have liked/ love to be orange!'	fox + dolphin	dolphin
	8.4	1.46	Θα μου άρεσε να ήμουν πράσινο! 'I would have liked/ love to be green!'	crocodile + kangaro	kangaro
	8.5	1.47	Θα μου άρεσε να ήμουν ροζ! 'I would have liked/ love to be pink!'	flamingo + giraffe	giraffe
	8.6	1.48	Θα μου άρεσε να ήμουν κόκκινο! 'I would have liked/ love to be red!'	crab + swan	swan

Appendix C: Colourful Creature Figures

Figure C1 Yellow creature

Figure C2 Red creature

Figure C3 Purple creature

Figure C4 Pink creature

Figure C5 Orange creature

Figure C6 Blue creature

Figure C7 Brown creature

Figure C8 Grey creature

Figure C9 Green creature

Appendix D: Animal Figures

Figure D1 Cat

Figure D2 Pig

Figure D3 Colibri

Figure D4 Giraffe

Figure D5 Horse

Figure D6 Kangaroo

Figure D7 Fox

Figure D8 Elephant

Figure D9 Turtle

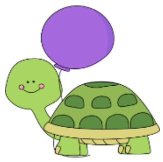

Figure D10 Crocodile

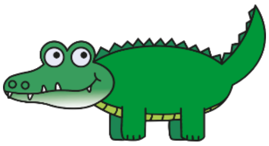

Figure D11 Flamingo

Figure D12 Whale

Figure D13 Parrot

Figure D14 Dolphin

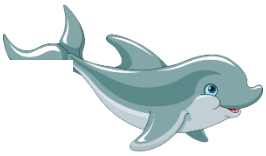

Figure D15 Swan

Figure D16 Tiger

Figure D17 Crab

Figure D18 Dog

Figure D19 Snake

Figure D20 Hippo

Figure D21 Penguin